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Executive Summary

We initiate coverage on BYD (002594.SZ) with a buy recommendation and a 12-month price target of ¥320.3, reflecting a 14.1% premium over its December 23, 2024, closing price of ¥ 280.7. As the largest dedicated EV manufacturer, BYD is spearheading the shift toward sustainable mobility and green energy. With a focus on expanding its export capabilities and optimizing single-car profitability, BYD is positioned as a leader in the NEV sector. Its strong supply chain control and strategic global expansion make it a compelling investment opportunity.

Thesis #1 BYD's Technology Moat & Leadership is Underestimated

BYD has established a formidable technology moat through ongoing innovation and substantial R&D investments, totaling over 100 billion yuan with a robust patent portfolio and a large team of over 100,000 R&D engineers. Since acquiring Xi'an Qinchuan Automobile in 2003, BYD has focused on plug-in hybrid technology, leading the market with groundbreaking innovations in new energy vehicles and expanding its technology system across electrification, super hybrid platforms, and intelligent layouts. The company dominates the domestic market, selling over 3 million vehicles in 2023—five times more than Tesla. Looking ahead, BYD is projected to sell over 5 million vehicles in China, capturing a 20-25% market share. The company's vertical integration strategy enhances its supply chain control, covering upstream raw materials, midstream production, and downstream sales and services, which stabilizes resources and mitigates supply chain disruptions.

Thesis #2: BYD's Improved Capital Efficiency & Profitability is Underestimated

BYD's capital efficiency and profitability benefit from scale and cost advantages. The peak of its plant expansion is over, with depreciation pressure expected to decrease. Recent capacity expansion included new projects in Changzhou, Jiangxi Fuzhou, Hefei Changfeng, and Xi'an, with a total domestic capacity of 4.18 million units. Depreciation expenses are declining, improving operational efficiency. The price war's impact has minimized, and BYD's vehicles now offer a significant price advantage over fuel cars. The company's single-vehicle profit has increased from a slight profit to more than 10,000 yuan per vehicle due to scale effects. With production currently around 3 million units annually, BYD has room to boost profitability further as it approaches the critical scale for cost optimization.

Thesis #3: BYD's Expansive Product Range & Global Expansion Strategy are Underappreciated

BYD's product portfolio and global strategy cover all passenger car segments, with plans to launch over 13 new models in 2024, ranging from 80,000 to 1 million RMB, including SUVs and supercars. The company is advancing vehicle intelligence and aims to export 450,000 cars while locally producing 400,000 units. BYD's global R&D and manufacturing network across Europe, Asia, and America supports international growth. Key collaborations with partners like NVIDIA, and SIPC further its new energy and green technology initiatives.

Exhibit 1: Company Data

Target Price	¥320.3
Last Close	¥280.7
Upside	14.11%
Current Market Cap.	¥717.7 billion
52-week High	¥167.00
52-week Low	¥327.38
Shares Outstanding	2.9 billion
PE TTM	22.98

Source: Team Analysis

Exhibit 2: Valuation Results

Valuation Method	Price
DCF – Terminal Growth	¥320.3
Comps – P/E	¥170.66
Comps – EV/EBITDA	¥310.15
Comps – EV/Revenue	¥381.10
Weighted Industry P/E	176.67
Target Price	¥320.3
Implied Return	14.1%

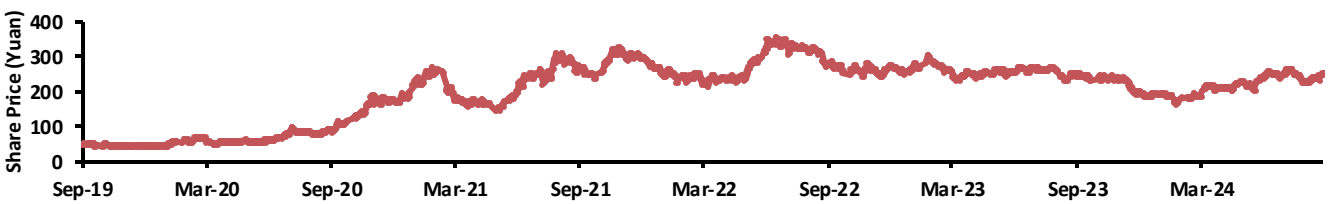
Source: Team Analysis

Exhibit 3: BYD Key Financial Data

	22A	23A	24E
Rev. (¥M)	424,061	602,315	837,217
Rev. Growth	96%	42%	39%
Adj. EBITDA (¥M)	41,523	78,141	104,034
Adj. EBITDA Growth	101%	88%	33%
ROE	15%	20%	20%
P/E	75.5x	16.5x	13.8X

Source: Company Filings/Team Analysis

Exhibit 4: BYD 5-Year Stock Chart



Source: Wind

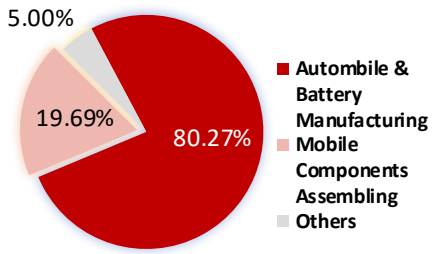
Business Overview

Founded in February 1995 and headquartered in Shenzhen, BYD is a leading high-tech enterprise with a global presence spanning over 30 industrial parks across 6 continents. With over 20 years of rapid growth, BYD has strategically positioned itself in the electronics, automotive, new energy, and rail transit sectors, playing a critical role in advancing zero-emission solutions. As a dual-listed company in Hong Kong and Shenzhen, BYD achieved a turnover and total market value exceeding 100 billion yuan as of FY24, driven by its commitment to technological innovation and enhancing quality of life.

Key Operation Segment

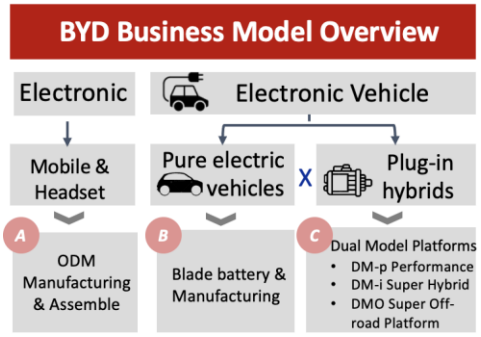
BYD generated over 60 million revenue in FY23, which could be divided into 3 business segments. 1) Automobile & Related Products. In terms of Automobiles, BYD mainly sells full electric vehicle and plug-in hybrid car. According to the sales revealed in FY23, BYD sold around 1.57m full electric vehicles in FY23-24, which accounts for 52.27% sales of all its electric cars (**Exhibit 5**). Meanwhile, BYD also supplies other electric bands with its "blade battery" as a critical component of electric vehicle in this segments. 2) Mobile Handset Components, Assembly Service and Other products. Another major business of BYD is assembling electronic device and components. Other than Foxconn, BYD is one of the most famous foundry in China, whose major clients including Apple, Samsung, etc.

Exhibit 5: BYD FY23 Key Business Segment (Revenue)



Source: Company Filings, 2023

Exhibit 6: BYD Business Model Overview (Electronic & EV)



Source: Team Analysis

Industry Overview

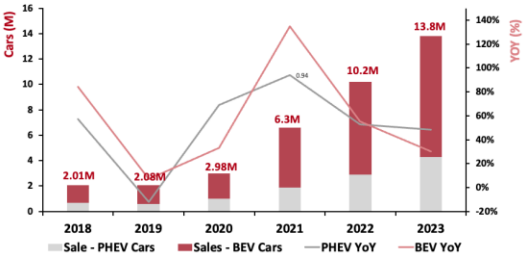
The Chinese new energy vehicle (NEV) market, led by pure electric vehicles (BEVs) with a growing share of plug-in hybrids (PHEVs), saw significant sales in the first five months of FY24, reflecting evolving consumer preferences and market capacity.

Global & China EV Market Capacity

On-going rationale expansion of global EV market. Electric car sales have sustained an upward trajectory and are projected to reach approximately 17 million units by the end of 2024. Between FY2018 and FY2023, the global EV market expanded significantly from 2 million to nearly 14 million vehicles **(Exhibit 7)**. This surge was particularly notable in 2021, fueled by favorable policy changes and a post-COVID-19 rebound. However, following this period, the EV market has transitioned to more rational growth patterns. In 2023, the electric car segment saw battery electric vehicles (BEVs) achieve a 48.2% year-over-year (YoY) increase in sales, while plug-in hybrid vehicles (PHEVs) experienced a 30% YoY rise.

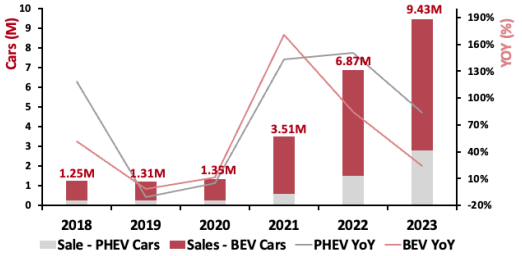
Critical Shifts in the Chinese EV Market. The global EV sales are predominantly concentrated in a few key markets, with China leading the charge. According to the International Energy Agency (IEA), over 60% of global new EV registrations in 2023 were in China, accounting for around 10 million sales of EV in 2023 **(Exhibit 8).** The primary driver behind the global market's growth has been China's rapid expansion. Notably, in 2023, China achieved significant growth without the support of national subsidies. The Chinese EV market is now entering a phase characterized by intense price competition and market consolidation.

Exhibit 7: Global EV Market Sales of PHEV & BEV Car from FY18 - FY23



Source: China Automobile Association, 2023

Exhibit 8: China EV Market Sales of PHEV & BEV Car from FY18 - FY23



Source: China Automobile Association, 2023

Major Trends & Shifts in EV Market

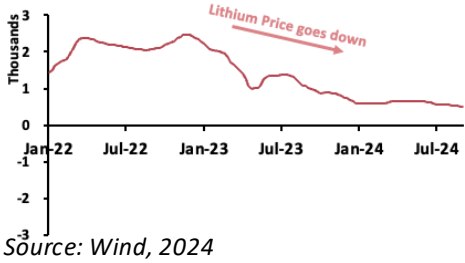
Positive – Reduction in Lithium Raw Material Prices. From 2022 to 2024, the price of lithium for power batteries has steadily decreased, as evidenced by the China Lithium Battery Price Index (LPI), which dropped from approximately 2000 on January 22, 2022, to 500 by September 1, 2024 **(Exhibit 9)**. Power batteries, a major component of electric vehicles (EVs) accounting for 40% to 50% of production costs, are likely to continue experiencing price reductions.

This trend is beneficial for EV manufacturers such as BYD, as it significantly lowers production costs. Additionally, the reduction in lithium prices enhances the value sustainability of EVs by allowing for lower market prices. Consequently, this could lead to increased profit margins and higher sales volumes for EVs.

Positive – Consistent Export Growth of EVs and BEVs in China. Despite economic recession and declining consumer spending, China's EV exports have remained robust, reaching approximately 5.2 million units in 2023 and accounting for 41.14% of the total cars exported **(Exhibit 10).** In response to sanctions from the U.S. and Europe, Chinese EV enterprises have adopted the strategy of *"Local factory + Brand direct operation"* to reduce production costs and avoid Tariffs Reasonably. Furthermore, Chinese EV companies are benefiting from supportive policies in Brazil, Thailand and Malaysia, which are aiding in expanding their market share beyond China. This consistent export growth not only underscores the resilience of China's EV sector but also positions companies like BYD to capitalize on global market opportunities.

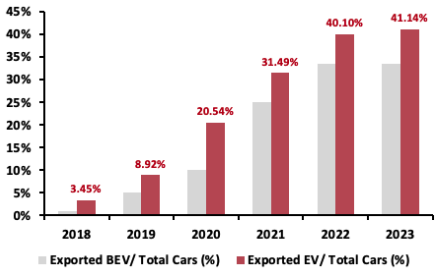
Positive – Accelerated high-end market development in China. In 2023, NEV sales in the RMB 300,000+ segment reached 929 thousand units in China, representing a YoY growth of 121%, with market share increasing from 12% in 2022 to 22%. Unlike the low profitability often seen in lower-priced NEVs, high-end NEVs are achieving gross margins of 15%-20%, significantly improving the overall profitability. The mid- to high-end market is primarily driven by robust demand for SUVs and MPVs, which are increasingly comparable to or even surpassing traditional luxury internal combustion engine vehicles in terms of comfort, and intelligent features.

Exhibit 9: China Lithium Battery Raw Material Price Index (LPI) From FY21 – FY24



Source: Wind, 2024

Exhibit 10: Annual Percentage of Exported EV in China from FY18 - FY23



Source: The General Administration of Customs of China, 2023

Competitive Positioning

BYD adopts the kangaroo theory as its development strategy, with decomposition of Self-running Stage, Breeding Bag Stage and the Long Leg Stage.

Self-Running Stage

Vertical integration of an entire industry chain. Compared to those competitors in the industry, BYD has its unique vehicle self-developed capability, including all the core upstream and downstream components **(Exhibit 13).** By contrast, most of its competitors choose to outsource most of components, which bring several uncertainties, for instance, the slow progress of mass production of multiple parts such as batteries provided by suppliers leads to a delay in the delivery of vehicles. The comprehensive full industry chain ensures BYD's self-developed and self-produced, enabling a stable production capacity to have mass production and reducing amortization of single automobile, greatly mitigating BYD's costs.

Breeding Bag Stage

Comprehensive product matrix. BYD releases nearly 20 new cars and forms a well-established product matrix, covering nearly all price ranges and especially dominating in the mainstream price range of 100 to 300 thousand yuan **(Exhibit 14).** Additionally, the series of YANGWANG was released in FY23, exploring the high-end area with an eye-watering price of 1.09 million yuan. Among the matrix, the Ocean series draws on the successful experience of existing models in parameter settings to form a strong product force while Destroyer05 and Frigate07 draws on experiences of the Dynasty series to reduce development costs. The combination of these brands in different segmented market make significant contributions to BYD's overall sales performance, achieving 31.9% of the market proportion in FY2023 with a year-on-year growth of 4.8%.

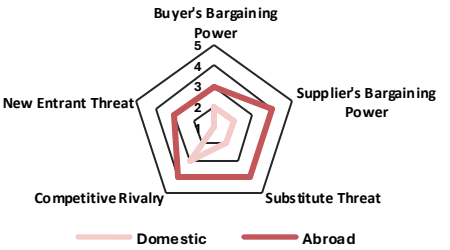
Adopting multi-industry Strategy. From the brand interaction of automobile, BYD also plays a role in the production of mobile phone components, assembly, and related products, occupying 19.68% of the total revenue.

Exhibit 11: Oversea Sale Operation Methods Comparison

Criteria	Direct Export	Local Manufacturing
No Tariff Barrier	✗	✓
Policy Support	✗	✓
Potential Brand Image	✗	✓

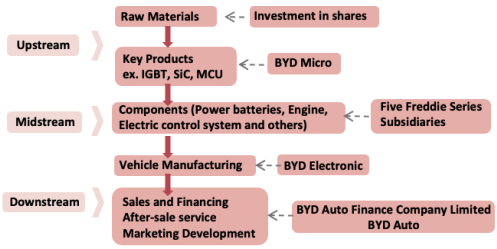
Source: Team Analysis

Exhibit 12: BYD Porter's Five Forces Analysis Matrix



Source: Team Analysis

Exhibit 13: BYD's Vertical Integration Chain



Source: Company Filings, 2023

The Long Leg Stage

Innovation-Driven Leadership in NEV Market. During FY22-23, BYD's investment in R&D development increased with a rate of 97%, covering 6.63% of its operating revenue **(Exhibit 15),** which also stood above all the competitors in the field. At present, BYD has 11 research institutes and has applied for over 40000 patents worldwide, with over 28000 authorized patents obtained. The disruptive technologies launched by the company, such as blade batteries, DM-i super hybrid and e-platform 3.0 have injected new vitality into the development of the new energy vehicle industry. Apart from that, its series of plug-in hybrid products occupied such a dominant share of 36.2% in FY23 that only Ideal ONE and Wenjie M5 can approach Tang and Han respectively, while no other car company can pose a threat to BYD, demonstrated in **Exhibit 17.** Plug in hybrids are BYD's most core market, which is the main driving force behind the company's continued sales lead over Tesla.

Investment Summary

Thesis #1: BYD's Technology Moat & Leadership is Underestimated by Market

1.1 BYD has established a strong technology moat through continuous innovation and significant R&D investments. Since acquiring Xi'an Qinchuan Automobile in 2003, BYD has focused on plug-in hybrid tech and introduced several innovative techs in 2008, leading in new energy vehicles. BYD continues to enhance its technologies and has invested over ￥100 billion in R&D showed in **Appendix 1**, employing over 100,000 engineers and holding numerous patents, which strengthen BYD technology moat in the market.

1.2 BYD has absolute advantage in the domestic selling markets. In 2023, BYD sold over 3 million cars in China, five times more than Tesla. As EV will replace gasoline cars eventually, BYD is expected to sell over 5 million vehicles in China, capturing 20-25% of the market and becoming the top domestic seller **(Exhibit 18).**

1.3 BYD's vertical integration strategy covers the entire EV industry chain, providing strong supply chain control. With internal investment and supply and external partnerships, BYD mitigates supply chain volatility and enhances brand loyalty through a robust sales and service to enhance its position in the market.

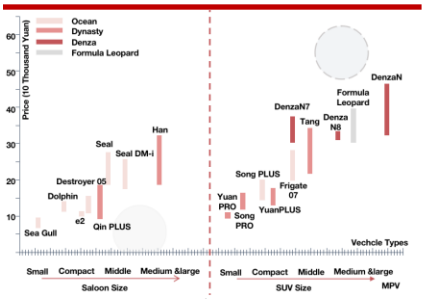
Thesis #2: BYD Capital Efficiency and Profitability has been underestimate by the market

2.1 For the capital expenditure, the apex of the firm's plant construction and expansion has passed. In 2022, BYD rapidly expanded production capacity, constructing factories and reaching a planned domestic capacity of 4.18 million units. Construction-in-progress account has declined recently, showed in **Exhibit 19,** reflecting depreciation burdens will decrease, improving operating efficiency.

2.2 The price war has basically bottomed out, and the scale effect of cost reduction brings short-term unit profit inflection point. BYD offers a price advantage over fuel cars, with limited need for further price cuts. The Qin PLUS glory version, priced at ￥104,800, offers significant savings compared to fuel cars, and orders exceeded expectations compare with same class **(Appendix 2).**

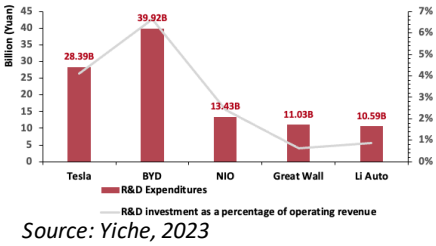
2.3 Historically, the single-vehicle profit is consistent with the trend of sales volume, and the scale improvement pushes the single-vehicle profit to increase. The Maxcy-Silberston Curve in **Exhibit 19** shows that production scale effects lead to cost reductions, with a minimum effective size around 5-7 million vehicles. BYD, currently at 3 million vehicles per year, has room to enhance profitability.

Exhibit 14: BYD's Main Products between 100 to 300 Thousand Yuan



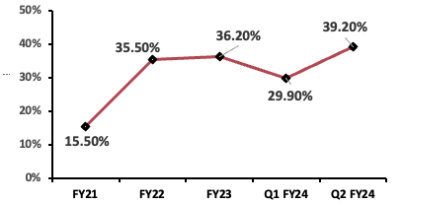
Source: Company Filings, 2023

Exhibit 15: R&D Expenditure among EV Companies in FY23



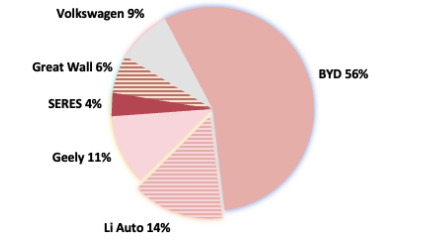
Source: Yiche, 2023

Exhibit 16: BYD's Share of Plug-in-Hybrid



Source: CPCA

Exhibit 17: Plug-in Hybrid Market in FY23

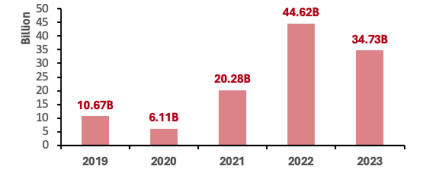


Source: CPCA

Exhibit 18: BYD Sales Estimation for FY24– FY26

Source: Company Filings, 2023

Exhibit 19: Construction Work in Process from FY19 to FY23



Source: Wind

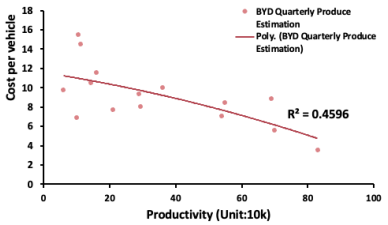
Thesis #3: BYD's Broad Product Range & Global Expansion Strategy are undervalued

3.1 The company has gradually realized the coverage of all categories and price segments of passenger cars. It now covers all price range of new energy vehicles and plans to launch over thirteen new models in 2024, including SUVs around ¥ 200,000 and supercars, addressing previous gaps in its lineup, and managing to cover buyers from all segments.

3.2 BYD continuous development of intelligence of vehicles has accelerated the global layout of factories. BYD has built a comprehensive manufacturing and R&D network across major continents **(Exhibit 20),** while Marketlines predicts BYD will export 450,000 cars and produce 400,000 locally in 2024 **(Exhibit 21).**

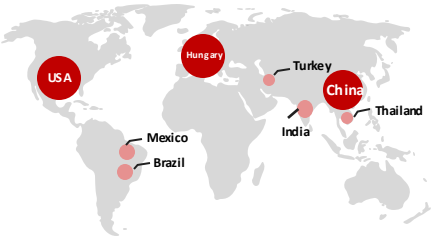
3.3 BYD has continued to strengthen its cooperation with global partners. It has deepened cooperation with NVIDIA for smarter cars, signed a strategic agreement with Weichai Power to advance new energy commercial vehicles, and partnered with State Power Investment Corporation about energy issues.

Exhibit 19: BYD Quarterly Sales Scale Effect



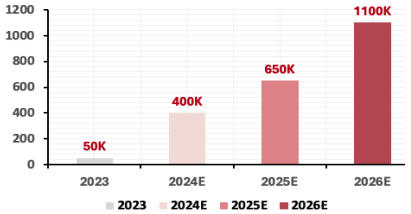
Source: Markline, 2023

Exhibit 20: BYD Oversea Production Factories in FY23



Source: Company Filings, 2023

Exhibit 21: BYD Oversea Local Productivity from FY22 to FY25E



Source: Wind

Valuation

We issue a buy recommendation on BYD with a target price of ¥320.3 (Exhibit 25), which represents 14.1% premium of the closing price of BYD on 2024/12/23 (¥280.7). Our recommendation is based on the Discounted Free Cash Flow Model (DCF) (**Appendix 5**) and relative valuation model including P/E, EV/Revenue and EV/EBITDA (**Exhibit 22**). We don't assign weight to the relative valuation result and only use results for cross-validation since the number of comparable are small and may not be representative enough.

Discounted Cash Flow Model (DCF)

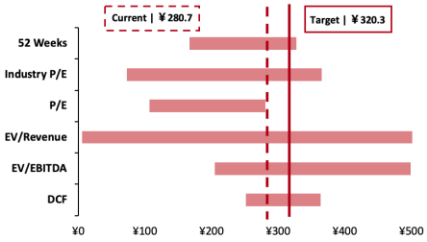
Transportation Manufacture and Related Revenue [CAGR of 16.1%]: This segment mainly compromised of automobile business, railway transit business, and battery business. Benefited by the post-pandemic recovery of global auto market, continuous development and innovation, and the political inclination to the electrical vehicles. The sales of BYD's Dynasty Series, Ocean Series, and DENZA Series were all greatly boosted, leading to a promising growth in the revenue in past few years. But due to the market saturation and limited capacity, the growth rate in FY24 will be estimated as 50% and grow at CAGR of 16.1% in next 7 years.

Consumer and Electronics and Related Revenue [CAGR of 12.5%]: This segment primarily focus on consumer electronics, phone components and assemble business. However, the global smartphone and PC markets are still facing economic challenges, with global shipment growth slowing down, limiting business growth in this segment, predicting that the growth rate will be 25% in FY24 and grow at CAGR of 12.5% till FY30.

Other Business Revenue [CAGR of -5.4%]: By considering the historical change of revenue (with annual decline of 35% with 2 consecutive fiscal years) in this segment and the possible challenges regarding to the other business BYD might face in the future market. BYD might continue to gradually reduce and contract the operation in other business sectors and shift more strategy and development focus to the main business sectors (**Exhibit 23**). Thus, the revenue will increase at a CAGR of -5.4%.

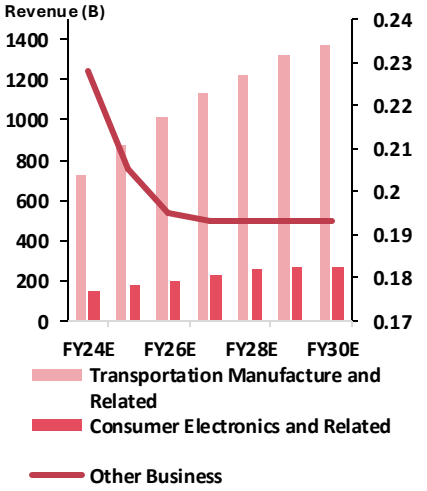
OPEX [CAGR of 15.5%]: The operating expense of BYD mainly composed of raw material cost and manpower salary, R&D expense, sales and marketing expense, general management expense, and business tax. The raw material and manpower salary increase dramatically in past 4 fiscal years largely due to the quick expansion of BYD's main business, since the majority of BYD's suppliers are located in China, the raw material expense's change will remain relatively constant. We predict the OPEX will keep increasing at a rate of 61% in FY24 but decline sharply after FY24 due to the internal business adjustment after years of expansion.

Exhibit 22: Valuation Football Field



Source: Team Analysis

Exhibit 23: Estimated Revenue by Segments (Billion)



Source: Team Analysis

WACC

BYD's WACC is 8.38% The exact method of calculating BYD's WACC is provided in **Exhibit 26**.

Cost of Debt [Before-tax: 4.31%]: The cost of debt is calculated by determining the respective interest rate of short-term debt and long-term debt. And compute weighted average of interest debt (26% of ST-Debt and 74% of LT-Debt) to acquire the before-tax cost of debt, and by taking tax into account, the eventual after-tax cost of debt is 3.23%.

Cost of Equity [9.23%]: By determining the factors in CAPM model, we calculated the Cost of Equity of BYD based on levered beta. Unlevered beta is calculated based on the regression test between the daily rate of return on BYD's stock and daily rate of return on CSI300 Index over the past 5 years.

- **Risk-free rate [2.24%]:** The risk-free rate is determined by 3-months average rate return of China 20 years national bonds from 2024/05 to 2024/08.
- **Market Risk Premium [5.26%]:** The market risk premium is calculated by subtracting the risk-free rate from the rate of return on CSI300 index in last 5 years.
- **Levered Beta [1.33]:** Unlevered beta (1.07) is calculated through the security characteristic line model and levered beta is computed based on the tax rate and debt-to-equity ratio of BYD.

Terminal Growth Rate

We assume after 2030, the growth rate of BYD's FCF will align with the growth rate of economies' GDP. Since BYD operate in a wide range of countries/regions (markets), 68% of it's revenue comes from China (including Hong Kong, Macao, and Taiwan) and 32% from oversea market. We calculated revenue weighted average of GDP growth (**Exhibit 29**) of those markets and the terminal growth rate is 3.25%.

Sensitivity Analysis

In order to test the robustness of our DCF models, we conduct a sensitivity analysis on WACC and Terminal Growth Rate (TGR). A +/- 1% on WACC represent the bullish or bearish perception on BYD's overall risk from perspective of different investors。Similarly, a +/- 0.5% on TGR reflects the possible fluctuation on the global economic strength where 2.75% is set as lower limit and 3.75% is set as upper limit as we don't believe global economy will fall neither into great recession nor great boom in the long run.

The test result (Exhibit 27) shows that the implied share price is the most sensitive to change in WACC, a 50-bps change will lead to a +38%/-25.1% change from original implied share price, 56.9% upside or 14.6% downside of share price given 3.25% of TGR. The change caused by sensitivity will drag stock price down however, even in the worst case where WACC is 9.38% and TGR is 2.75%, the implied share price is 217.3, 22.6% downside the current stock price, but still higher than BYD's 52 weeks low. Nevertheless, this case is unlikely to occur given the risk resistance of BYD and world inflation rate. Therefore, the DCF model illustrate a relatively high robustness.

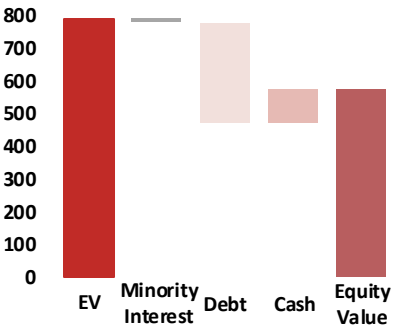
Scenario Analysis

Next, we conduct scenario analysis by identifying 3 different scenarios, Bull, Base, and Bear. For each scenario, we vary multiple model assumptions listed in (**Appendix 6**). Where in the bull case, the target price is 353.9, 26.1% upside from the current closing price while in the bear case, the target price is 251.18, 10.5% downside the current closing price (**Exhibit 28**).

Relative Valuation

We conduct relative valuation on BYD including EV/EBITDA, EV/Revenue, and both comparable companies' P/E and industry P/E since the number of comparable is limited. The 9.9x EV/EBITDA multiple suggest a 310.15 valuation and 1.5x EV/Revenue generate a 381.10 valuation (**Appendix 8**). For P/E valuation using Industry P/E, we choose the P/E of automobiles industry, electronic component industry and power battery industry and calculated the weighted average industrial P/E of 17.6x by assigning the 70% weight to automobile industry, 20% to electronic components industry and 10% to power battery industry.

Exhibit 24: Base Case Waterfall (Billion)



Source: Company Filings, 2023

Exhibit 25: DCF Output (Million)

Terminal Value	1140798
PV of Terminal Value	649479
PV of Forecast Period	143869
Enterprise Value	**793348**
Less: Minority Interest	11652
Less: Net Debt	201548
Equity Value	**580148**
Shares Outstanding (A)	1811.3
Implied Stock Price	**320.3**

Source: Team Analysis

Exhibit 26: WACC Buildup

Pre-tax Cost of Debt	4.31%
Corporate Tax Rate	25.00%
After-tax Cost of Debt	**3.23%**
Risk-free Rate	2.24%
Market Risk Premium	5.26%
Levered Beta	1.33
Cost of Equity	**9.23%**
Net Debt Value	10,653,687
Market Value of Equity	64,764,886
WACC	**8.38%**

Source: Team Analysis

Exhibit 27: Sensitivity Analysis

WACC (%)	Terminal Growth Rate (%)				
	2.75	3.0	3.25	3.5	3.75
7.38	387.2	412.3	440.5	472.4	508.6
7.88	332.1	351.9	378.8	398.3	425.8
8.38	286.9	302.8	320.3	339.6	360.9
8.88	249.2	262.2	276.4	291.9	308.9
9.38	217.3	228.1	239.8	252.5	266.3

Source: Team Analysis

Exhibit 28: Scenario Analysis

	Bear	Base	Bull
Revenue CAGR	10.5%	15.4%	18.7%
EBITDA Margin	10.4%	11.7%	12.8%
FCFF CAGR	22.7%	24.8%	27.2%
WACC	8.38%	8.38%	8.38%
TGR	3.0%	3.25%	3.5%
Share Price	251.18	320.30	353.89
Premium	-10.5%	14.1%	26.1%

Source: Team Analysis

The result suggests a max range of 265.2 to 364.9 valuation (**Appendix 9**). However, the results are not suggestive enough since those companies lack the business similarities (firm size, market share, industry range) and financial similarities (profitability, financial ratios) with BYD. Therefore, we mainly focus on the result provided by DCF model.

Financial Analysis

Anticipate Strong Profitability Margin

In FY21, the gross profit margin of the company decreased from 18% to 11.6% due to an increase in the cost of original materials. With the new energy compensation of government, the net profit margin in FY21 maintained positive signals that is 1.8%. Despite this, the gross profit margin began to increase back from 11.6% to 18% annually post-2021, supported by the decline of materials price, advanced technology and more comprehensive product metrics. EBITDA margin, NOPAT margin, and net profit margin mirrored the trend of the gross profit margin from 1.4% to 4.8% and from 1.8% to 5.2%. What's more, the decrease in EBITDA in FY21 was also due to **higher development and research expenses** in blade battery and intelligent technology to improve the safety of e-car. Additionally, the net profit margin fell in FY21 as pandemic-related tax credits ended. Through a thorough examination of the key profitability ratios with other **competitors** like SERES and Great Wall Motor, we can deduce the following findings: Firstly, BYD's advanced technology and comprehensive metrics enable it to maintain a higher profit margin compared to its competitors in a highly competitive industry; Additionally, the EBITDA margin indicates that BYD has a slight edge in profitability compared to other companies in the industry, showed in **Exhibit 36.**

Solvency Capability Recover Gradually

Current ratio: The current ratio of BYD in FY22-24 is around 0.7, which is lower than the level of the industry 0.9, which indicates that BYD has a weak ability to repay current liabilities with the cash generated from the realization of current assets. However, the pressure of cash flow is understandable which is influenced by the increase of inventory in a short period, and it will go back to a normal level in the near future **(Exhibit 31).** **Quick ratio:** The drop from 76% to 44% in the quick ratio in FY22 is due to the increase of accounts payables caused by the much **slower payment process (Exhibit 33).** **Cash ratio:** BYD's ratio which is around 20% is slightly lower than that of Great Wall Motor's 40%, which indicates that BYD has a **higher level of inventory,** resulting in funds in inventory, there is difficulty in realizing the cash quickly, as well as slower return of accounts receivable **(Exhibit 32).**

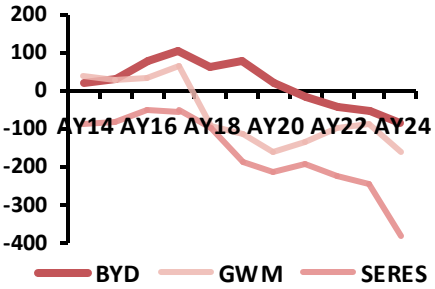
Turnover Features Stabilize in The Near Future

Working capital to sales ratio maintains a negative trend in the recent decade since BYD's current assets are dramatically lower than its current liabilities which shows that they have a high level of non-current assets, such as Property, Plant, and Equipment. The **PPE ratio** is keeping about 2.5 which means that BYD invested too much capital and many funds to develop and build their new factories and get the property in many domestic areas and foreign countries. These factories help BYD to escape the higher tax rate of foreign countries and build their own comprehensive product lines which contains each parts of the e-car, such as battery, intelligent screen and high technology. The increase of **Inventory turnover and trade receivable turnover** from FY19's 5 and 2.8 to FY23's 6.87 and 9.32 indicates that BYD paid more attention to trade receivables collection capabilities and manage the inventory. The fluctuating trend in trade receivable turnover could be due to changes in credit policies or sales strategies, and the accounts receivable turnover in FY22 was much higher than other years, such as longer credit terms during the difficult period of the pandemic in FY22. The turnover rate of BYD's inventory and trade receivables was lower than that of Great Wall Motor's , indicating that BYD had the problem of excessive inventory hoarding and long collection time.

Exhibit 29: GDP Growth (Regions)

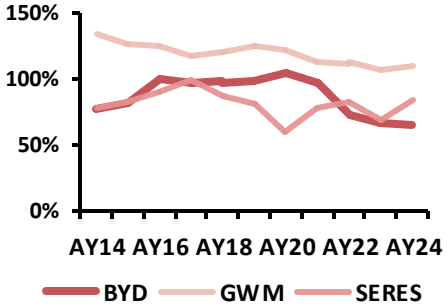
Source: Wind

Exhibit 30: Cash Conversion Cycle from AY14 – AY24



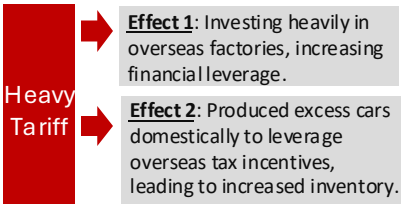
Source: Team Analysis

Exhibit 31: Current Ratio From AY14 – AY24



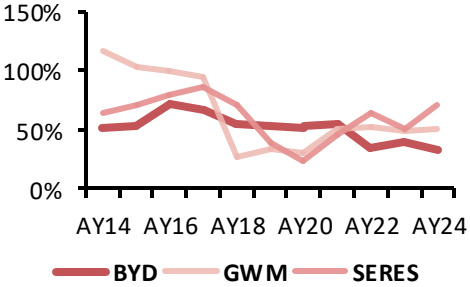
Source: Team Analysis

Exhibit 32: Exporting Tariff Effect on BYD Financial Performance



Source: Team Analysis

Exhibit 33: Quick Ratio From AY14 – AY24

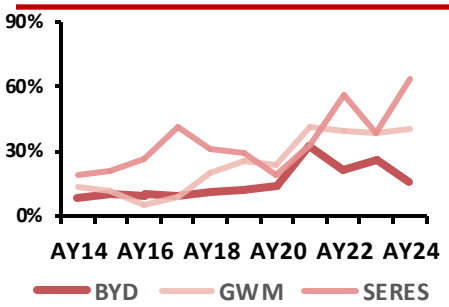


Source: Team Analysis

Excellent Cash Flow Management Capability after Pandemic

Cash conversion cycle was negative from FY21's -18 to FY24's -86, which indicates that BYD paid its suppliers after receiving cash from customers, implying strong cash flow manage capability. The huge decrease from 21 to -18 in FY21 is attributed to the increase in BYD's business in FY21, including a substantial decrease in the trade receivable amount and increase in the account payable amount. At the same time, due to the **investment on suppliers, expansion of factories, creating new sub-companies which support the key materials and manage the financial activities to achieve the vertical integration of the whole supply chain**, BYD has a **strong bargaining power** of material suppliers and did not have to pay cash to suppliers quickly, which led to the increase of Days' payable showed in **Exhibit 30**.

Exhibit 34: Cash Ratio from AY14 – AY24

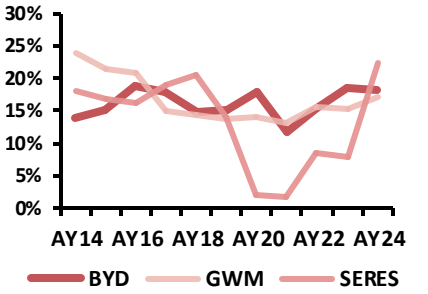


Source: Team Analysis

DuPont Analysis: Heavy Financial Leverage

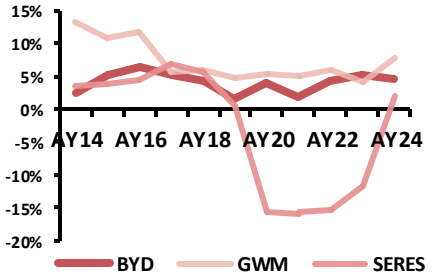
From FY21 to FY23, the **Return on Equity (ROE)** showed in **Exhibit 37** consistently increased from 2.8% to 19.8%, indicating continuous improvement in profit generation. The net operating profit margin also rose annually from 1.4% to 4.8%, reflecting a growing share of operating profit in total profit. This growth was fueled by the post-pandemic recovery of consumer activities and support policies and regulations of new energy industry. During the past five years, **operating asset turnover** kept increasing from 2 to 9.6 but saw a decline which is 8.8 in FY23 due to several factors. The competitive and homogeneous nature of the new energy industry drove down the price per car in FY23 as the company traded higher volume for lower unit prices, **increasing overall income but reducing per-unit asset revenue.** Secondly, an increase in **international tariffs and tax rates** such as 100% tariffs in Canada and 36.3% in Europe caused a significant decrease in operating income from the company's supply chain and international business. **Financial leverage** also rose each year from 0.24 to 0.59, with a significant decrease between FY20 and FY21, driven by repaying a large amount of long-term loans and payables. Despite BYD's more effective use of financial leverage compared to Great Wall Motor suggesting a more reasonable debt structure, its overall financial leverage was higher than the industry average across the three years, indicating higher financial risk which BYD should pay more attention to **(Appendix 16).**

Exhibit 35: Gross Profit from AY14 – AY24



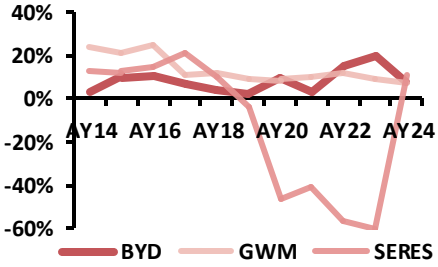
Source: Team Analysis

Exhibit 36: Net Profit Margin from AY14 – AY24



Source: Team Analysis

Exhibit 37: ROE from AY14 – AY24



Source: Team Analysis

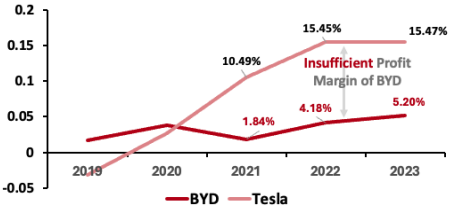
Investment Risks

Risk #1: Unideal Net Profit Margin on Sales

In the past years, BYD owns a relatively low Net Profit Margin on Sales with an average of 3.344%, reflecting a poor manage ability in profitability. What leads investors even worried more is Tesla, holding an average Net Profit Margin on Sales of 8.198% and with a higher overall ROE in 2023 **(Exhibit 38).**

Mitigate: BYD's risk resilience is strengthened by its diverse product range and expanding overseas operations. The introduction of the luxury SUV U8, priced at $213,000 with a 30% profit margin, is expected to enhance BYD's Net Profit Margin on Sales and counteract sales shortfalls. Additionally, BYD's focus on high-end products has contributed to a Net Profit Margin on Sales of 3.5% in Q1 2024, with a recent peak of 3.82%, indicating improved profitability.

Exhibit 38: Net Profit Margin on Sales for BYD & Tesla from FY19 – FY23



Source: Team Analysis

Risk #2: Market Competition Risk

With the intensifying competition in the new energy vehicle market, technological upgrades and market expansion by new entrants and existing firms may affect BYD's profitability. Last year, the overall industry maintaining a growth rate of 180% in sale, BYD only increase for around 69%, showing a fierce competition market.

Mitigate: Though the EV industry is already a red sea market, BYD are putting more fund into R&D to maintain the tech advantage it already owns, while also seeking for capturing new market by investing more in marketing fee showed in **Exhibit 39**, Such endeavor reflecting in the figures of BYD sales overseas, which maintaining a CAGR of 50.79% since 2014, while also improve the reputation and recognition of the corporation.

Risk #3: Market Expansion Risk

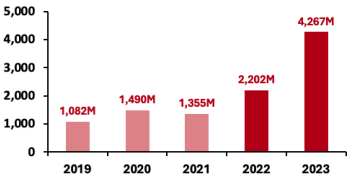
Mentioned in BYD's annual report, the company plans to focus on expanding its business by exporting cars to other countries. However, since 2024, the EU has imposed tariffs on Chinese EVs for various reasons. BYD faces a tariff of around 27% per car, posing a significant challenge for BYD's export efforts. Additionally, Brazil, another major market for BYD, has implemented a 10% tax on vehicle exports. Given the volatile nature of political relations between countries, BYD encounter further tariff adjustments in other markets **(Exhibit 41).**

Mitigate: Mentioning in their annual report, BYD actively promotes the localization of production processes, driving the global transformation towards automotive electrification. In December 2023, BYD has announced plans to establish a manufacturing base in European Hungary, actively promoting technological exchanges and innovation between China and Hungary, further deepening its market presence in Europe.

Risk #4: Chips Provision Vulnerability

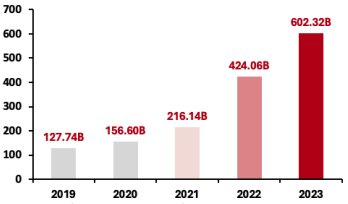
The shift towards advanced electronic architectures in EVs has made chip supply crucial for systems like intelligent driving and smart cockpits. BYD relies on expensive chips from NVIDIA due to limited suppliers meeting their standards. The tense China-US relations increase the risk of chip shortages, potentially disrupting BYD's production and competitiveness.

Exhibit 39: BYD Marketing Expenditures from FY19 – FY23



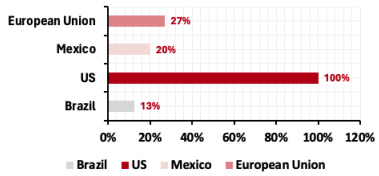
Source: Company Filings

Exhibit 40: BYD Oversea Sales from FY19 – FY23



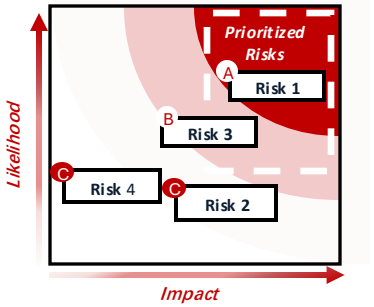
Source: Choice, Wind

Exhibit 41: Tariff in Major BYD's Exporting Nations



Source: Ministry of Commerce of the PRC

Exhibit 42: Investment Risk Exposure Matrix Evaluation



Source: Team Analysis

ESG

BYD has a specific CSR Management Committee to coordinate all parties involved to ensure its favorable ESG performance.

Environment Perspective

Carbon Emissions. BYD always insists on reducing carbon emissions, gradually establishing zero carbon parks as early as FY21. It became the world's first automaker to cease the production of gasoline powered vehicles in FY22, leading the industry's green transformation. At the same time, BYD fully leverages its unique advantages in the field of new energy and builds the first "zero carbon park" for Chinese automotive brands at its global headquarters in Shenzhen, which fully utilizes green solutions such as photovoltaics, energy storage and new energy vehicle to reduce carbon dioxide emissions by a total of 245000 tons. At the COP28 "China Corner" Shenzhen Special Session held in 2023, BYD's "Zero Carbon Park" won the top ten green and low-carbon cases in Shenzhen. During the latest reporting year, BYD added 93 energy-saving transformation projects for production technology and process management, such as building energy and carbon management platforms, with a total energy-saving benefit of 17160 tons of standard coal and a reduction of 132490.83 tons of CO_2 emissions.

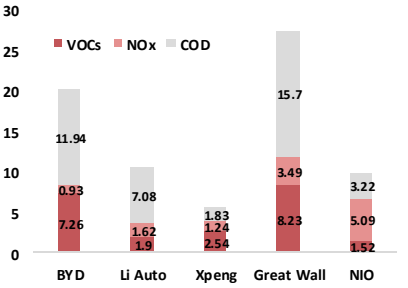
Pollutant Emissions. In terms of air pollutants, BYD's volatile organic compounds (VOCs) emissions in 2023 were 7.26g per ten thousand yuan with a year-on-year decrease of 37.41% **(Exhibit 44).** In terms of water pollutants, BYD's ammonia nitrogen (NH3-N) emissions in 2023 reached a year-on-year decrease of 4.1%, but its chemical oxygen demand (COD) emissions gained a year-on-year increase of 12.11%. Considering BYD's booming sales and its rather adorable price, this fluctuations are understandable and it's undeniable that the company made significant progress in controlling pollutant emissions in FY23.

Exhibit 43: ESG Scores of Peers

	Rating	Environment	Society	Governance
BYD	A	9.3	3.6	4.6
Great Wall	A	5.2	3.9	5.1
Li Auto	AAA	7.8	4.6	5.5
Xpeng	AAA	8.6	4.7	6.1
Tesla	BBB	9.2	1.4	3.0

Source: MSCI

Exhibit 44: Peer Pollutant Emissions Density (g/RMB10,000revenue)



Source: Company Files

Social Perspective

A Diverse and Inclusive Workplace. The globalization process simultaneously buttresses BYD to present the humanity against the background of multi-cultures. During FY21-23, BYD remains a high level in female workforce engagement, occupying 32% of the total employees and 14% of the senior management team in FY23 **(Exhibit 45).** Compared to its peers, BYD captures a leading place in its gender diversity. Besides, acting as a technology manufacturing enterprise, R&D personnel account for 15% of total staffs **(Exhibit 46),** maintaining a nearly 50% year-to-year increase rate .

Safe Workplace and Labor Management. In FY23, BYD set up its Safe Factory star rating system for the first time, establishing exhaustive safety audit standards to measure the maturity of its safe environmental protection, occupational health and fire management levels in each factory with quantitative standards, improving the factorial security management. Admittedly, the LDR has seen an obvious decline to 0.69, compared to 3.03 and 1.48 in FY20 and 22 respectively **(Exhibit 47).** Apart from that, BYD also provides a clear and transparent salary growth and profit-sharing mechanism with increasing paid leave off and a series of employee care campaigns, resulting in up to 60% approval which represent the percent of staff who are willing to recommend the enterprise.

Downsides in Consumer Satisfaction. BYD's performance might be weakened by the high complaint rate about car quality, turning out that seven of the top ten new energy vehicle series that received the most complaints belonged to BYD, with Song series' 5018 times ranking the first during FY23 to Q1 FY24. Besides, Great Wall Motor sued BYD for non-compliance with vehicle evaporation pollution emissions standards in FY23, which was an adverse public sentiment, resulting in a considerable customer trust. However, we still hold unswerving trust on it since BYD has established a comprehensive product traceability system and fast complaint channels to avoid product liability risks and respond to the problems immediately. In addition, the company has established strict defect car recall control procedures to reduce the adverse effects caused by quality control of automobiles and problem product recycling. In FY23, BYD Auto did not have any recalls **(Exhibit 48).**

Governance Perspective

Executive Remuneration. BYD's director rights are held under successive Long Term Incentive Plans (LTIP) as the company completed equity incentives during a early period. For instance, Wang Chuanfu, as the actual controller, directly holds 513 million shares and the vice president Mr. Lyu obtains 239 million shares in FY23. The market value of the stock they hold are far higher than their salary value, which is closely related to the corporate performance. Based on this, we firmly believe that they will work harder to achieve a higher firm value.

Shareholder Ownership. BYD adopts a stable natural person holding structure with the founder Wang Chuanfu as the actual controller, who owns a total shareholding of 17.64% in Q1 FY24 **(Exhibit 49)**, and the remaining natural person shareholders are all company executives. It's clear that the company's shareholding structure is relatively decentralized, which can easily lead to the risk of the separation of ownership and operation. However, a certain degree of decentralization enable an adequate employee equity incentives system. Besides, during FY23, Mr. Yang Dongsheng, the vice president of the company and Ms. Zhou Yalin, the vice president and chief financial officer of BYD increased their shareholding of 18600 and 13500 shares respectively, unfolding a robust signal of management's conviction in BYD's future development.

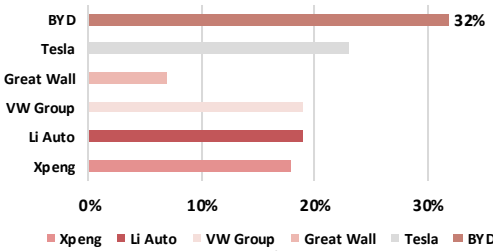
Board of Directors. Wang Chuanfu has been BYD's CEO for the past 22 years, slightly higher than the industry standard while all of the directors in the BOD also share a tenure of 22 years except for the new joiner, indicating that a consistent and solid management team can lead BYD to achieve a higher grade and overcome the difficulties. It's also notable that in FY23, one female director succeeded to join the board, breaking the deadlock that without woman, increasing its board diversity. However, the proportion of female directors in BYD is still lower than the industry average.

Exhibit 45: BYD Staff Statistics

	Percentage of male	Percentage of female	Percentage of male in senior position	Percentage of female in senior position	Percentage of R&D personnel
FY19	67%	33%	83.40%	16.60%	9.60%
FY20	66%	34%	83.10%	16.90%	15.95%
FY21	67%	33%	86%	14%	14%
FY22	69.50%	30.50%	87%	13%	12.20%
FY23	68%	32%	85%	15%	15%

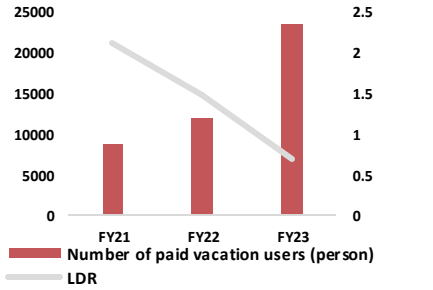
Source: Company Files

Exhibit 46: Peer Female Staff Ratio



Source: Company Files

Exhibit 47: Labor Management



Source: Company Files

Exhibit 48: Product Liability

Product Liability	FY21	FY22	FY23
Percentage of Products Recalled for Safety & Health Reasons	3.10%	3.36%	0

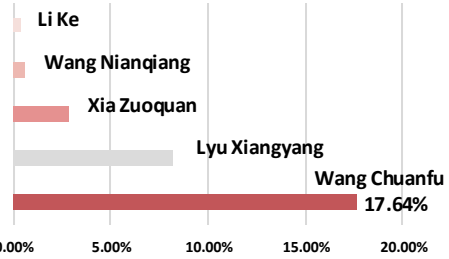
Source: Company Files

Exhibit 49: Company's Approval Rate

	Company Rating	CEO Approval Rate
BYD	3.5	64%
Li auto	3.7	100%
Tesla	3.5	58%
Great wall	3.2	26%
Xpeng	3.3	69%
VW Group	4.1	87%

Source: Glassdoor

Exhibit 49: Top5 Inside Individual Shareholders



Source: Wind

Appendix

1. BYD R&D Expenditure in FY14– FY23
2. Average Selling Price of BYD and Competitors in FY23
3. Historical Share Price Performance VS. New Vehicle Index
4. Annotated share price history
5. Base case discounted cash flow
6. Assumptions about DCF Model (Base, Bull, Bear)
7. Stock Price in Different Scenarios
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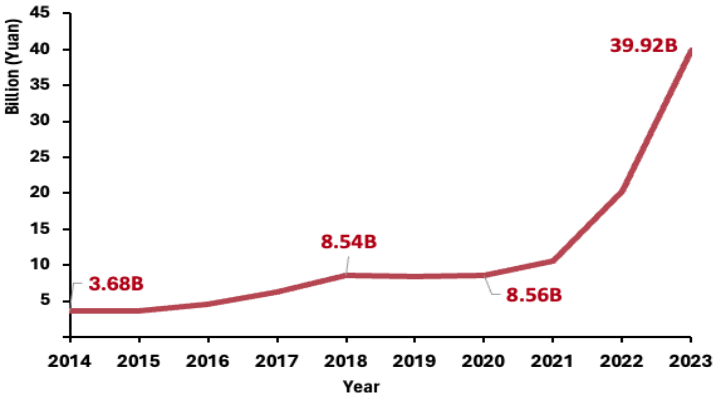
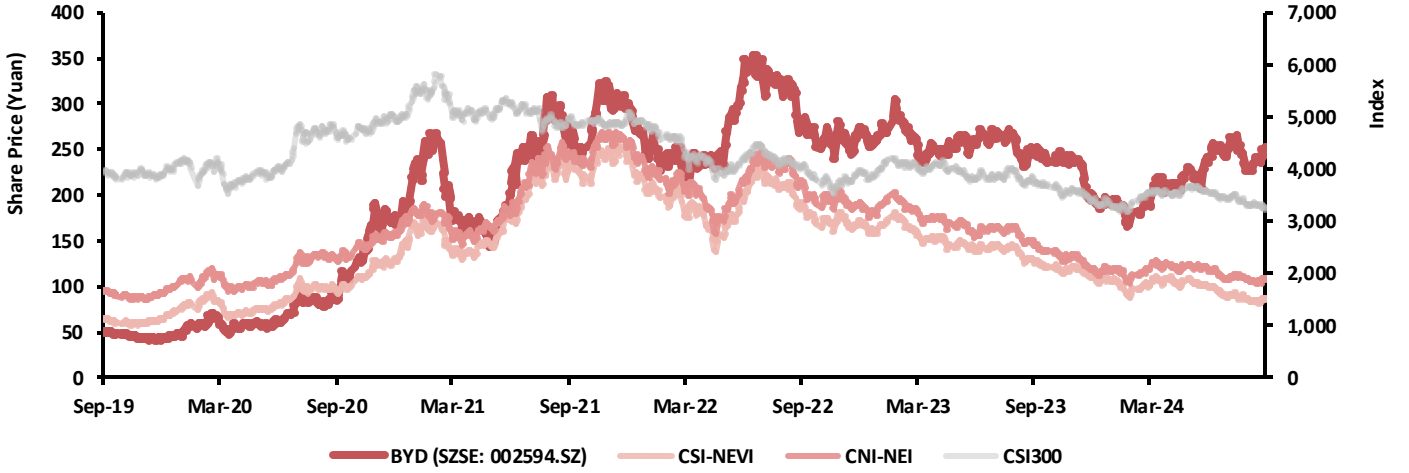
Appendix 1: BYD's R&D Expenditure in FY14– FY23

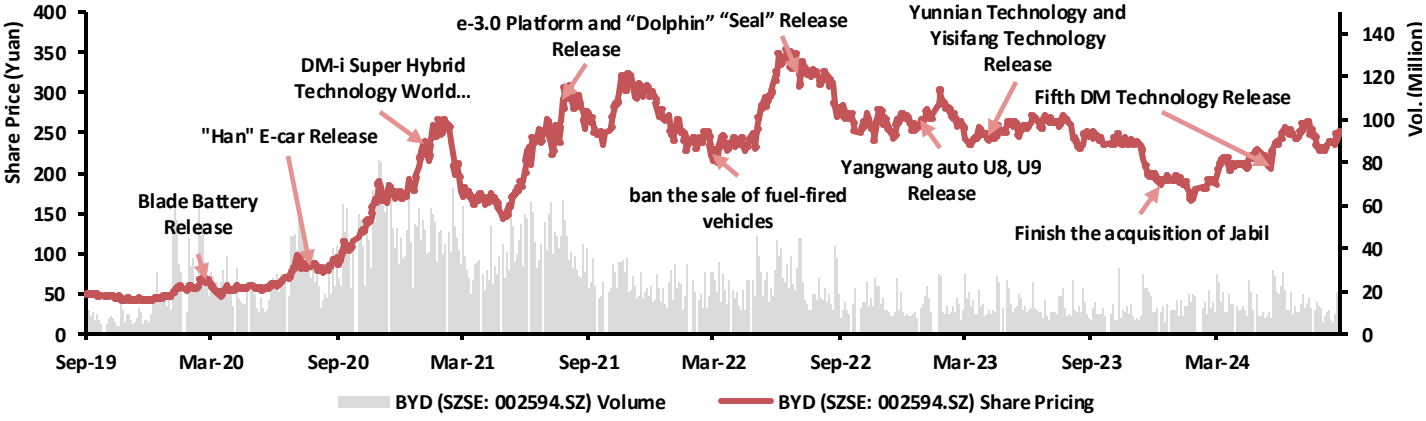


Appendix 2: Average Selling Price of BYD and Competitors in FY23

Appendix 3: Historical Share Price Performance VS. New Vehicle Index



Appendix 4: Annotated Share Price History



Appendix 5: Base Case Discounted Cash Flow

Fiscal Year Ended	Historical				Forecast						
million RMB	2020A	2021A	2022A	2023A	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Revenue	**156597.7**	**216142.4**	**424060.6**	**602315.4**	**873629.2**	**1052574.0**	**1220942.7**	**1363338.4**	**1474723.2**	**1585056.2**	**1640439.7**
Transportation Manufacture and Related	83993.3	112489.2	324691.2	483453.3	725180.0	877467.8	1017862.6	1129827.5	1220213.7	1317830.8	1370544.0
Consumer Electronics and Related	60043.0	86454.5	98815.1	118576.9	148221.1	174900.9	202885.1	233317.9	254316.5	267032.3	269702.6
Other Business	473.9	727.5	554.4	285.1	228.1	205.3	195.0	193.1	193.1	193.1	193.1
COGS	**126251.4**	**187997.7**	**351815.7**	**480558.4**	**701524.3**	**842059.2**	**975533.2**	**1066130.7**	**1160607.2**	**1244269.1**	**1283644.1**
Gross Profit	30346.3	28144.7	72245.0	121757.0	172105.0	210514.8	245409.5	297207.8	314116.1	340787.1	356795.6
Gross Margin	19.4%	13.0%	17.0%	20.2%	19.7%	20.0%	20.1%	21.8%	21.3%	21.5%	21.8%
OPEX	18996.4	22817.7	50989.6	88597.7	142793.5	162559.0	188561.8	207826.6	226280.8	234625.0	242823.1
EBITDA	**24112.7**	**20609.8**	**41523.0**	**78141.0**	**98112.9**	**125626.0**	**147025.1**	**162479.5**	**167109.1**	**191367.4**	**185736.3**
EBITDA Margin	15.4%	9.5%	9.8%	13.0%	11.2%	11.9%	12.0%	11.9%	11.3%	12.1%	11.3%
EBIT	11594.0	6501.4	21153.0	34588.3	32590.7	51945.8	61559.2	94312.5	93372.9	112114.6	120118.7
Tax	868.6	550.7	3366.6	5924.6	3993.0	6364.3	7542.1	11555.0	11439.9	13736.1	14716.8
NOPAT	**10725.4**	**5950.7**	**17786.3**	**28663.8**	**28597.7**	**45581.5**	**54017.0**	**82757.5**	**81933.0**	**98378.5**	**105401.9**
(+) D&A	12518.7	14108.4	20370.0	43552.7	65522.2	73680.2	85466.0	68166.9	73736.2	79252.8	65617.6
(-) CAPEX	(11774.1)	(37343.6)	(97456.9)	(122093.5)	(147643.3)	(177885.0)	(180699.5)	(163600.6)	(162219.6)	(158505.6)	(130743.0)
(-) Change in NWC	6236.8	-10368.2	-87347.3	-59004.2	-65522.2	-73680.2	-61047.1	-40900.2	-44241.7	-23775.8	-16404.4
UFCF	5233.2	-6916.4	28046.8	9127.1	11998.7	15056.8	19830.6	28223.9	37691.3	42901.5	56680.8

Enterprise Value	793347.8
(-) Minority Interest	11652.0
(-) Net Debt	201547.8
Market Value of Equity	580148.0
Number of Shares Outstanding (Diluted)	1811.3
Implied Share Price	**320.3**

Appendix 6: Assumptions about DCF Model (Base, Bull, Bear)

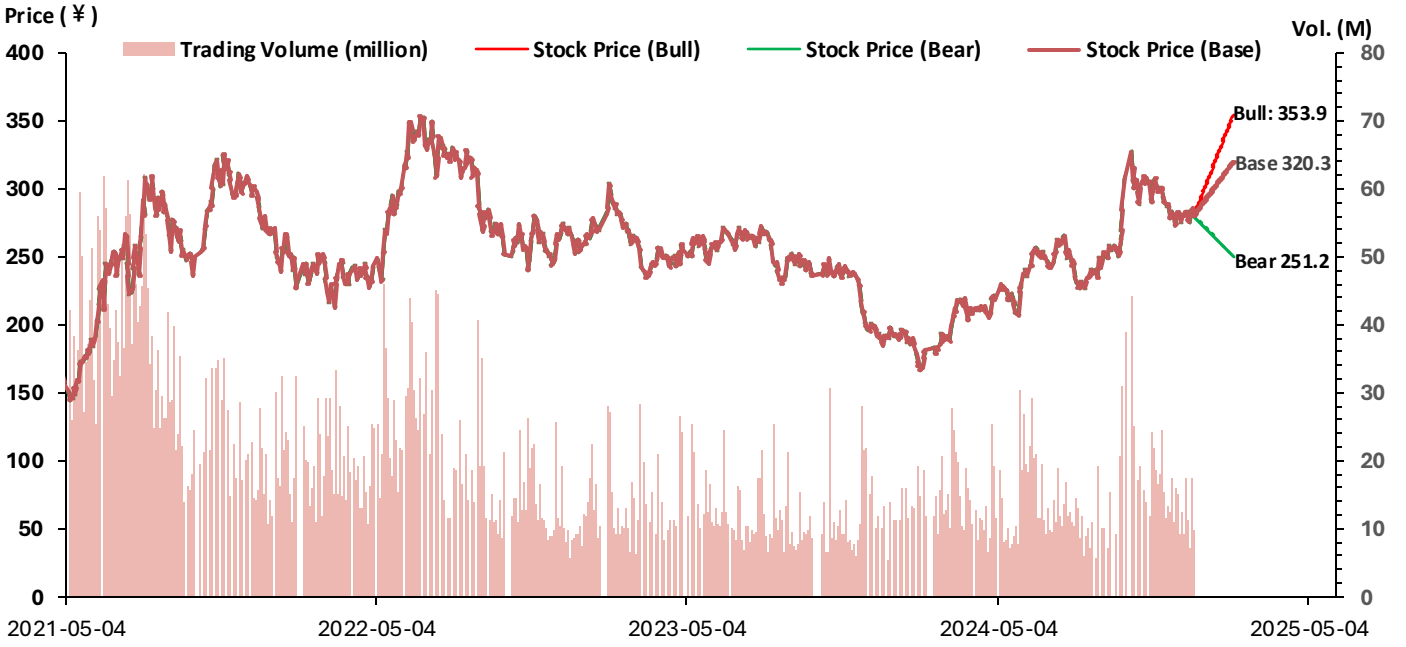
Base Case	Assumptions
Revenue	
Transportation Manufacture and Related	By looking at the past trend on the revenue growth in this segment and the number of EV sold in the past, we forecast that the revenue will continue to increase due to the driving factors: ·policy support·breakthrough innovation on critical technologies of EV such as rechargeable battery, high-performance chassis, power unit, body control system and advanced driving assistance system. ·Demand around the globe ·Customers awareness shift. However, there are also hindrance: ·Trade regulation and tariff increase regarding to exported vehicles. ·Environmental regulation toward the battery recycle and disposal ·Increasing competition in the market ·Lack of intelligent driving system ·Market saturation. Therefore we assume the revenue will remain at the growth rate of FY23 in FY24 (around 50%) and then experience big fall in FY25 and gradually slow down in future years.
Consumer Electronics and Related	By examining the global consumer electronics and predicting the shipment quantity. The global consumer electronic market will recover from the winter and benefited by the manufacture in relatively new emerging economies with low-cost labor forces. The market will be more prosperous than several years ago. Thus lead to the growth rate of revenue in this segment increase from 20% in FY23 to 25% in FY24, and thus remain for 1 year and then goes down.
Other Business	By looking at the proportion of each segment's revenue in the total revenue, it's easy to find the proportion of other business' revenue is less than 1% of total revenue and continue to decrease from FY20 to FY23. BYD is increasing it's profitability by shifting the resource and strategic focus to the auto related business from others and we assume BYD will keep contracting it's investment on other less core business. Thus the other business' revenue will decrease at the geometric average of decline rate (10%) in past 5 years, and stop decreasing in FY28.
COGS	The regression test shows growth rate of BYD's total COGS is align with the growth rate of company's total revenue. Most of it's COGS comes from the purchase of raw materials, production related matters and the salary to the manpower. Based on the previous assumption for revenue segment. The COGS will grow at the rate accordingly.

OPEX	
Sales & Marketing Expense	The sales & marketing expense remain relatively constant at a percent of revenue, so we assume the sales & marketing expense will hold at the average percent of revenue in the future, which is 3.4% plus 1% extra for it's possible expansion into new oversea market.
R&D Expense	Since BYD increase it's R&D investment from 3.7% to 6.6% of it's revenue in past 3 fiscal years largely due to the quick expansion strategy. Based on the financial success in recent years, we assume BYD will keep it's R&D investment at 6.6% of it's revenue in FY24 and FY25 and then remain at average of 5.3% in following years.
G&A Expense	Relatively stable in the past, so we assume it will remain at the average of 3.5% of revenue in the future.
Business Tax	Stable in the past, but consider the possible contract in Chinese tax policy, the business tax will be assumed at 2% of it's revenue, 0.5% upside from it's historical average.
Non-Cash Item	
D&A	Estimated to be at the historical average, 7.5% in FY24 but slightly decrease in following years as the purchase of fixed asset or the M&A activities will slow down in the future.
CAPEX	Estimated CAPEX to be the average of 17% of revenue in FY24 but will slowly goes down in the following years as the business becomes more mature in domestic market and focus more on oversea market.

Bull Case	Assumptions
Transportation Manufacture and Related	BYD accelerate the innovation and avoid environmental regulation risk to EV by launching more advanced models, increasing the battery disposal technologies. Both domestic and oversea market improve their favor toward BYD's auto products. Thus lead to a jump in CAGR of it's revenue from 16.1% in base case to 20% in bull case.
TGR	Assume China and other major economic entities enter into the cyclical prosperity after 2030, thus increase TGR by 25bps to 3.5%.
COGS	Assume the raw material cost slightly fluctuate, lead an increase from 79.6% of revenue to 79.1% of revenue.
R&D	Increase the R&D expense to 7% of the revenue in FY24 and remain at an average of 6.2% in following years.
CAPEX	Increase CAPEX from 12.8% of revenue to 13.2% of revenue on average in forecasting period from FY24 to FY30 since BYD will expand to more oversea market to meet the increasing demand.

Bear Case	Assumptions
Transportation Manufacture and Related	Global demand for EV lower than the expectation, the sales of product lacks drivers. Geopolitical risk further reduces the expansion on oversea market. Causing a fall of CAGR of revenue from 16.1% to 10.0%.
TGR	Assume China and other major economic entities further slows down the economic growth rate after 2030, thus decrease TGR by 25 bps to 3%
COGS	Assume the raw material cost greatly fluctuate, lead an increase from 79.6% of revenue to 81% of revenue.
R&D	The R&D expense will slowly decrease in FY24 and FY25 since the lag of reacting to market, and R&D expense will fall down to 4.7% of revenue for the following years.
CAPEX	Decrease CAPEX from 12.8% of revenue to 11.2% of revenue since BYD may consider shrinking the expansion project.

Appendix 7: Stock Price in Different Scenarios



Appendix 8: Relative Valuation by Competitors

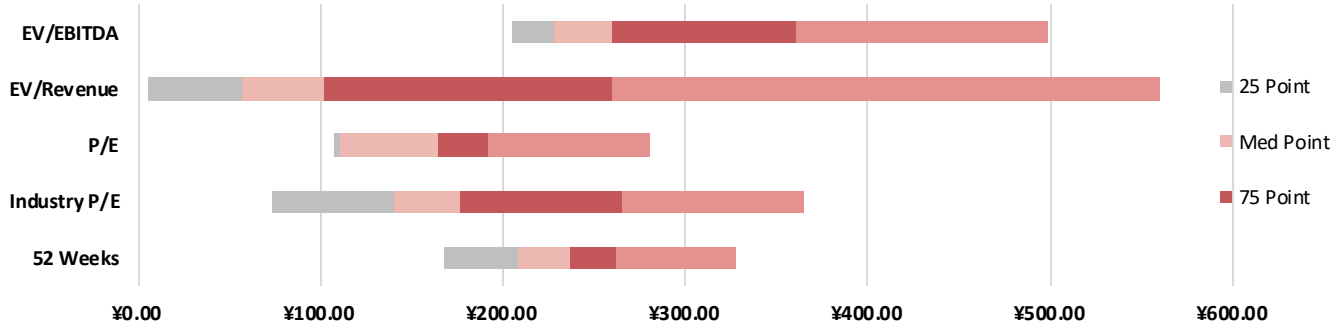
By Competitors	Market Cap (mn)	Enterprise Value (mn)	EV/EBITDA		EV/Revenue		P/E	
			2023	2024E	2023	2024E	2023	2024E
Great Wall Motor Co. Ltd.	138,846.08	196,814.21	14.3 x	10.2 x	1.1 x	0.9 x	27.2 x	21.7 x
SAIC Motor Co. Ltd.	145,270.01	276,191.13	8.7 x	6.7 x	0.4 x	0.4 x	10.4 x	8.4 x
Chana Auto	99,721.87	99,721.87	8 x	8.1 x	0.7 x	0.5 x	10.7 x	10.5 x
Geely Automobile	81,654.59	94,221.34	7.5 x	7.0 x	0.5 x	0.4 x	15.9 x	13.5 x
CATL	817,958.20	817,958.20	11.1 x	9.6 x	2.0 x	2.0 x	18.5 x	15.0 x
XPENG INC.	48,401.45	132,563.99	N/A	N/A	4.3 x	3.0 x	N/A	N/A
Average			9.9 x	8.3 x	1.5 x	1.2 x	16.5 x	13.8 x
Min			7.5 x	6.7 x	0.4 x	0.4 x	10.4 x	8.4 x
Median			8.7 x	8.1 x	0.9 x	0.7 x	15.9 x	13.5 x
Max			14.3 x	10.2 x	4.3 x	3.0 x	27.2 x	21.7 x

Relative Valuation	EV/EBITDA	EV/Revenue	P/E
2023 Multiples	9.9 x	1.5 x	16.5 x
Enterprise Value	682610.48	684387.73	
Market Cap	580334.33	582111.59	
Implied Price	310.15	381.10	170.66

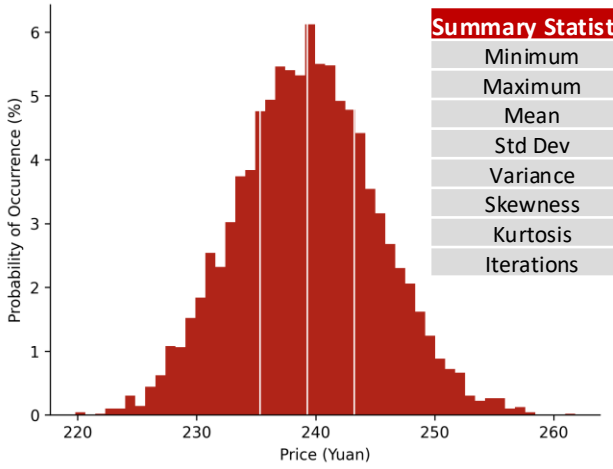
Appendix 9: Relative Valuation by Industry

By Industry Segments	Min	25th	Median	75th	Max	BYD EPS (2023)	Weight	Min	25th	Median	75th	Max
2023 Electronic component industry P/E	5.8 x	11.6 x	15.5 x	19.7 x	29.3 x	10.32	20%	59.65	119.97	160.06	203.36	302.79
2023 Power battery industry P/E	10.7 x	16.9 x	18.0 x	21.9 x	49.1 x	10.32	10%	110.73	174.49	186.02	225.60	506.61
2023 Automobile industry P/E	6.9 x	13.7 x	17.5 x	28.0 x	35.1 x	10.32	70%	71.52	141.69	180.08	288.60	362.44
Weighted Average Price (70% to auto industry, 10% to power battery, 20% to electronic component)								73.07	140.63	176.67	265.25	364.93

Appendix 10: Distribution of Relative Valuation Results



Appendix 11: Monte Carlo Simulation



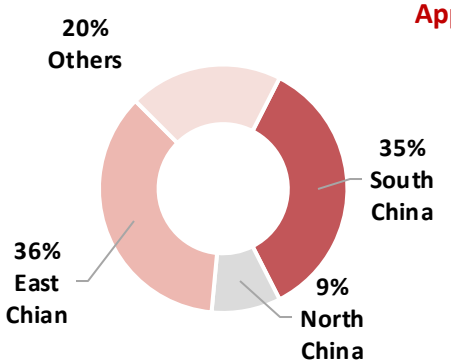
Summary Statistic	Value
Minimum	217.8053
Maximum	259.9809
Mean	239.4841
Std Dev	5.86893
Variance	34.44434
Skewness	0.052388
Kurtosis	-0.08971
Iterations	5000

Percentile	Output Price (Yuan)
1%	226.2697863
2.5%	228.2233006
5%	229.9891775
10%	231.9881442
20%	234.4292615
25%	235.4955365
30%	236.3645993
35%	237.1690467
40%	237.9153897
45%	238.6392268
50%	239.3847403
55%	240.071809
60%	240.9228956
65%	241.7377702
70%	242.5397865
75%	243.4104897
80%	244.4506513
90%	247.0078072
95%	249.2956275
97.5%	251.2029461
99%	253.2994358

Appendix 12: Car Sale Volume in China during FY23

No.	Brand Name	Growth Compare with 2022	2023 Sales	2022 Sales
1	BYD	50.34%	2,706,075	1,799,947
2	Tesla	37.27%	603,664	439,770
3	Aion	76.64%	483,632	273,757
4	Geely Automobile	53.96%	469,427	304,911
5	SGMW	3.56%	457,848	442,118
6	Chana Auto	81.32	384,915	212,277
7	Li Auto	182.2%	376,030	133,246
8	Great Wall Automobile	91.1%	236,856	123,920
9	NIO	1182%	160,038	12,486
10	Leapmotor	29.67%	144,155	111,168

Appendix 13: BYD's Suppliers Selection



BYD implements localized procurement and selects suppliers nearby. As of the end of 2023, BYD has more than 10000 cooperative suppliers, accounting for 35% in South China, 36% in East China, 9% in North China, and 20% in other regions. In recent years, due to the increase in demand and the expansion of business such as complete vehicles and batteries, BYD has set up factories in many places, and the proportion of suppliers in other regions keep on rising, driving local employment.

Appendix 14: BYD's Overseas Factories up to FY24

Factories	Planning production capacity (Ten thousand vehicles)	Other Information	Capacity coverage area
Uzbekistan's Jizak Oblast	30	Has been put into operation on January 25, 2024, covering an area of about 160 acres.	Central Asia region
Thai passenger car production base	15	Plan to complete and put into operation in the third quarter of 2024, covering an area of approximately 1500 acres	Surrounding ASEAN countries and other regions
Camasari, Brazil	15	Expected to be put into operation in the second half of 2024, covering an area of approximately 6900 acres	South America region
West Java Province, Indonesia	15	Construction begins in July 2024 and is planned to commence operations in 2026, covering an area of approximately 1500 acres	Surrounding ASEAN countries and other regions
Szeged City, Hungary	20	Construction is about to begin, and it is expected to be put into operation in 2024-2026	European region
Mexico factory (under site selection)	15	The factory will be located and constructed before the end of 2024, possibly in central or southern Mexico	North America

Appendix 15: BYD's SWOT Analysis in FY24

	Helpful To achieving the objective	Harmful To achieving the objective
Internal origin Attributes of the organization	**Close Industry Chain Connections:** A comprehensive industry chain layout ranging from upstream raw materials to midstream components and downstream vehicles, forming a complete closed-loop battery technology. **Advanced Technological Path:** The "Self-research + Cooperation" dual-track technological path enables the firm to maintain a leading position in these critical technologies while ensuring technological independence and safety. **Comprehensive Domestic Market Brand Layout:** By the end of 2023, BYD had over 1,400 dealerships in China, surpassing its competitors. Its five major brand matrices have achieved full market coverage, showing its diversification and deep layout in the new energy vehicle market.	**Brand Influence Needs Improvement:** BYD has long focused on producing lower-end models, and as a result, consumers have developed a strong, established perception of the brand. This "low-end" label has become deeply ingrained in the minds of many consumers. As a result, even though BYD has introduced premium models, their acceptance in the market may still be limited due to these existing brand perceptions.
External origin Attributes of the environment	**Supportive Incentive Policies:** For instance, subsidies for NEV purchases, tax incentives to encourage the growth of the NEV sector, and strong support for the development of NEV infrastructure. **Continued Expansion of the NEV Market:** Global attention on reducing greenhouse gas emissions and promoting sustainability increases, new energy vehicles have become a key development focus, which are regarded as substitutes for traditional gasoline-powered vehicles. **Ongoing "Going Global" Strategy:** With the rising global demand for new energy vehicles, China's foreign trade policies provide an opportunity for NEV companies like BYD to expand into international markets.	**Insufficient Social Demand:** Some domestic cities with underdeveloped charging infrastructure have limited market demand for new energy vehicles (NEVs) , with residents continuing to rely mainly on traditional gasoline-powered cars for daily transportation **Intense Competition from Strong Rivals:** BYD faces significant challenges in the new energy vehicle (NEV) market from a number of formidable competitors, including global leader Tesla, SAIC Group with its deep automotive manufacturing expertise, and other emerging domestic and international NEV brands.

Appendix 16: BYD and Competitors' Key Financial Ratios

Financial Ratios (BYD)

BYD	FY24	FY23	FY22	FY21	FY20	FY19	FY18	FY17	FY16	FY15	FY14
Return on equity	12.75%	19.81%	14.97%	2.84%	9.64%	2.00%	4.45%	7.07%	11.08%	10.40%	3.67%
Net operating profit margin	4.04%	4.84%	4.53%	1.44%	4.23%	1.45%	1.81%	4.00%	6.05%	2.70%	0.07%
× Operating asset turnover	4.42	6.29	6.80	4.85	2.13	1.93	1.93	1.69	1.73	1.90	1.58
= Return on Operating Assets	17.88%	30.44%	30.84%	6.97%	9.03%	2.79%	3.49%	6.77%	10.47%	5.14%	0.11%
Return on Operating Assets	17.88%	30.44%	30.84%	6.97%	9.03%	2.79%	3.49%	6.77%	10.47%	5.14%	0.11%
x (Operating Assets/Business Assets)	0.43	0.40	0.37	0.34	0.74	0.73	0.83	0.83	0.86	0.83	0.87
+ Return on Investment Assets	1.43%	1.38%	-0.21%	0.29%	0.03%	-2.03%	2.91%	0.49%	1.09%	18.79%	19.08%
x (Investment Assets/Business Assets)	0.57	0.60	0.63	0.66	0.26	0.27	0.17	0.17	0.14	0.17	0.13
= Return on Invested Capital (ROIC)	8.47%	13.01%	11.14%	2.59%	6.70%	1.47%	3.39%	5.68%	9.20%	7.50%	2.63%
Spread	7.46%	11.54%	9.45%	1.03%	5.48%	1.14%	3.04%	5.29%	7.57%	7.01%	2.22%
× Financial leverage	0.57	0.59	0.40	0.24	0.54	0.46	0.35	0.26	0.25	0.41	0.47
= Financial leverage gain	4.28%	6.80%	3.83%	0.25%	2.94%	0.52%	1.06%	1.39%	1.88%	2.90%	1.04%
ROE = Return on Business Assets + Financial leverage gain	12.75%	19.81%	14.97%	2.84%	9.64%	2.00%	4.45%	7.07%	11.08%	10.40%	3.67%
Line items as a percent of sales											
Revenue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Cost of sales	81.11%	81.50%	84.68%	88.38%	82.00%	84.93%	85.25%	82.24%	81.10%	84.72%	86.09%
Selling, general, and admin. expense	7.44%	6.18%	5.53%	6.28%	8.39%	9.00%	8.83%	13.24%	11.85%	12.16%	13.83%
Other operating income/expense	1.80%	0.87%	0.41%	1.05%	1.08%	1.35%	1.79%	1.18%	0.00%	0.00%	0.00%
Net operating profit before tax	4.04%	4.84%	4.53%	1.44%	4.23%	1.45%	1.81%	4.00%	6.05%	2.70%	0.07%
Net non-recurring income/expense	-0.95%	-0.26%	0.11%	-0.47%	0.13%	-0.68%	-2.23%	-1.10%	-0.45%	-0.62%	-0.46%
Investment and interest income	0.52%	0.39%	-0.06%	0.13%	0.01%	-0.46%	0.37%	0.07%	0.11%	2.39%	2.03%
Interest expense	0.23%	0.26%	0.23%	0.21%	0.31%	0.08%	0.07%	0.07%	0.25%	0.11%	0.11%
Tax expense	-1.01%	-0.98%	-0.79%	-0.25%	-0.55%	-0.24%	-0.64%	-0.66%	-1.05%	-0.82%	-0.23%
Profit or loss	5.23%	5.20%	4.18%	1.84%	3.84%	1.66%	4.30%	5.10%	6.39%	5.30%	2.28%
Operating expense line items as a percent of revenue (by function) (%)											
Cost of sales	81.11%	81.50%	84.68%	88.38%	82.00%	84.93%	85.25%	82.24%	81.10%	84.72%	86.09%
Selling, general, and administrative expense	7.44%	6.18%	5.53%	6.28%	8.39%	9.00%	8.83%	13.24%	11.85%	12.16%	13.83%
Key profitability ratios (%)											
Gross profit margin	18.895%	18.497%	15.323%	11.617%	18.003%	15.073%	14.751%	17.759%	18.901%	15.283%	13.909%
EBITDA margin	5.951%	6.055%	5.267%	2.169%	4.699%	2.443%	4.637%	5.761%	7.581%	3.836%	0.585%
NOPAT margin	4.042%	4.836%	4.532%	1.438%	4.234%	1.449%	1.813%	3.996%	6.055%	2.705%	0.072%
Net profit margin	5.226%	5.204%	4.177%	1.835%	3.840%	1.659%	4.301%	5.100%	6.390%	5.301%	2.281%
Turnover ratios											
Operating working capital/Sales	-45.59%	-37.94%	-34.06%	-27.59%	-4.10%	-11.38%	-3.18%	-3.53%	0.37%	-15.24%	-23.73%
Net non-current assets/Sales	68.19%	53.83%	48.75%	48.22%	50.98%	63.30%	55.07%	62.59%	57.47%	67.87%	87.01%
PP&E/Sales	57.863%	45.708%	42.362%	38.434%	39.363%	47.635%	41.031%	48.134%	44.883%	52.314%	66.586%
Operating working capital turnover	-2.19	-2.64	-2.94	-3.62	-24.41	-8.78	-31.47	-28.31	271.65	-6.56	-4.21
Net non-current asset turnover	1.47	1.86	2.05	2.07	1.96	1.58	1.82	1.60	1.74	1.47	1.15
PP&E turnover	1.73	2.19	2.36	2.60	2.54	2.10	2.44	2.08	2.23	1.91	1.50
Trade receivables turnover	6.09	9.32	10.41	5.74	3.70	2.81	2.59	1.77	2.12	2.78	2.46
Days' receivables	59.14	38.63	34.58	62.73	97.17	128.22	139.22	202.85	169.42	129.71	146.41
Inventories turnover	4.04	6.87	5.36	4.99	4.99	5.00	4.94	5.33	5.95	5.08	5.83
Days' inventories	86.68	50.95	65.29	70.20	70.17	70.07	70.86	65.67	58.78	68.90	60.01
Trade payables turnover	1.69	2.47	2.50	2.37	2.47	3.00	2.40	2.16	2.38	2.15	1.93
Days' payables	212.51	145.55	144.13	151.69	145.53	120.02	150.28	166.45	151.20	167.19	186.37
Cash conversion cycle	-66.69	-55.98	-44.26	-18.75	21.81	78.26	59.80	102.07	77.00	31.42	20.05
Liquidity ratios											
Current ratio	0.707	0.666	0.722	0.970	1.049	0.990	0.988	0.978	0.999	0.823	0.768
Quick ratio	0.340	0.404	0.338	0.547	0.533	0.539	0.543	0.663	0.720	0.536	0.530
Cash ratio	0.179	0.262	0.216	0.327	0.136	0.117	0.112	0.094	0.098	0.100	0.084

Financial Ratios (SERES)

SERES	FY24	FY23	FY22	FY21	FY20	FY19	FY18	FY17	FY16	FY15	FY14
Return on equity	11.26%	-60.53%	-56.02%	-40.30%	-45.91%	-3.38%	10.14%	21.55%	14.97%	12.48%	12.75%
Net operating profit margin	1.47%	-17.00%	-15.97%	-19.79%	-17.37%	-1.33%	2.89%	5.54%	3.58%	2.77%	2.99%
× Operating asset turnover	-2.14	-4.75	-3.61	4.54	5.92	5.16	3.39	8.36	14.41	3.99	4.05
= Return on Operating Assets	-3.15%	80.72%	57.65%	-89.83%	-102.76%	-6.88%	9.80%	46.29%	51.55%	11.06%	12.11%
Return on Operating Assets	-3.15%	80.72%	57.65%	-89.83%	-102.76%	-6.88%	9.80%	46.29%	51.55%	11.06%	12.11%
x (Operating Assets/Business Assets)	-2.14	-0.54	-0.62	0.27	0.24	0.26	0.51	0.27	0.19	0.63	0.68
+ Return on Investment Assets	0.15%	8.68%	-0.02%	2.23%	-0.94%	-0.18%	2.86%	0.71%	2.52%	7.09%	6.12%
x (Investment Assets/Business Assets)	3.14	1.54	1.62	0.73	0.76	0.74	0.49	0.73	0.81	0.37	0.32
= Return on Invested Capital (ROIC)	7.20%	-30.12%	-35.58%	-22.80%	-24.88%	-1.94%	6.38%	13.09%	11.95%	9.58%	10.17%
Spread	7.11%	-32.15%	-36.27%	-23.25%	-25.34%	-2.04%	5.92%	12.83%	10.88%	9.07%	9.82%
× Financial leverage	0.57	0.95	0.56	0.75	0.83	0.70	0.64	0.66	0.28	0.32	0.26
= Financial leverage gain	4.06%	-30.40%	-20.44%	-17.50%	-21.04%	-1.43%	3.76%	8.45%	3.02%	2.90%	2.58%
ROE = Return on Business Assets + Financial leverage gain	11.26%	-60.53%	-56.02%	-40.30%	-45.91%	-3.38%	10.14%	21.55%	14.97%	12.48%	12.75%
Line items as a percent of sales											
Revenue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Cost of sales	77.68%	92.14%	91.39%	98.33%	97.91%	85.80%	79.51%	81.01%	83.79%	83.25%	82.03%
Selling, general, and admin. expense	16.13%	19.82%	19.84%	15.63%	13.24%	14.54%	14.50%	12.28%	11.92%	13.52%	14.30%
Other operating income/expense	1.01%	1.53%	1.53%	1.76%	4.35%	2.91%	0.92%	0.39%	0.00%	0.00%	0.00%
Net operating profit before tax	1.47%	-17.00%	-15.97%	-19.79%	-17.37%	-1.33%	2.89%	5.54%	3.58%	2.77%	2.99%
Net non-recurring income/expense	-0.52%	-0.56%	-0.79%	-2.90%	-2.42%	-1.93%	-2.02%	-1.15%	-0.22%	-0.07%	-0.08%
Investment and interest income	0.13%	5.13%	-0.01%	1.31%	-0.51%	-0.30%	1.02%	0.28%	0.88%	1.23%	0.89%
Interest expense	0.01%	0.38%	0.11%	0.15%	0.15%	0.09%	0.13%	0.06%	0.10%	0.06%	0.03%
Tax expense	-0.52%	-0.21%	-0.85%	-0.13%	-0.33%	-0.92%	-1.39%	-1.45%	-0.89%	-0.63%	-0.85%
Profit or loss	2.09%	-11.60%	-15.31%	-15.74%	-15.62%	0.47%	5.62%	6.87%	4.45%	3.85%	3.78%
Operating expense line items as a percent of revenue (by function) (%)											
Cost of sales	77.68%	92.14%	91.39%	98.33%	97.91%	85.80%	79.51%	81.01%	83.79%	83.25%	82.03%
Selling, general, and administrative expense	16.13%	19.82%	19.84%	15.63%	13.24%	14.54%	14.50%	12.28%	11.92%	13.52%	14.30%
Key profitability ratios (%)											
Gross profit margin	22.320%	7.865%	8.607%	1.666%	2.086%	14.201%	20.487%	18.992%	16.209%	16.750%	17.968%
EBITDA margin	2.493%	-16.134%	-14.337%	-16.762%	-14.639%	1.772%	6.119%	8.095%	4.554%	3.309%	3.769%
NOPAT margin	1.471%	-16.995%	-15.975%	-19.793%	-17.371%	-1.333%	2.888%	5.540%	3.579%	2.770%	2.994%
Net profit margin	2.090%	-11.597%	-15.307%	-15.736%	-15.623%	0.471%	5.621%	6.866%	4.449%	3.851%	3.778%
Turnover ratios											
Operating working capital/Sales	-80.22%	-81.82%	-82.05%	-77.12%	-93.28%	-60.17%	-25.84%	-20.39%	-26.07%	-25.05%	-27.83%
Net non-current assets/Sales	33.47%	60.76%	54.34%	99.15%	110.18%	79.54%	55.30%	32.36%	33.01%	50.09%	52.55%
PP&E/Sales	18.149%	30.480%	29.221%	57.641%	66.859%	51.137%	38.128%	23.674%	23.242%	36.359%	36.858%
Operating working capital turnover	-1.25	-1.22	-1.22	-1.30	-1.07	-1.66	-3.87	-4.90	-3.84	-3.99	-3.59
Net non-current asset turnover	2.99	1.65	1.84	1.01	0.91	1.26	1.81	3.09	3.03	2.00	1.90
PP&E turnover	5.51	3.28	3.42	1.73	1.50	1.96	2.62	4.22	4.30	2.75	2.71
Trade receivables turnover	14.99	7.65	12.26	6.41	20.02	10.41	3.44	3.34	2.17	2.54	2.72
Days' receivables	24.02	47.04	29.38	56.15	17.98	34.57	104.59	107.84	166.16	141.80	132.23
Inventories turnover	18.36	10.16	8.54	6.03	7.25	8.06	11.42	16.39	14.84	14.69	11.58
Days' inventories	19.07	34.46	40.98	58.05	48.30	43.42	30.66	21.35	23.59	23.82	30.21
Trade payables turnover	0.85	1.10	1.21	1.18	1.28	1.36	1.56	1.98	1.49	1.46	1.44
Days' payables	422.91	328.68	296.48	304.63	281.16	264.13	230.86	181.77	240.97	246.00	249.41
Cash conversion cycle	-379.82	-247.18	-226.13	-190.43	-214.88	-186.14	-95.61	-52.58	-51.22	-80.38	-86.96
Liquidity ratios											
Current ratio	0.846	0.695	0.834	0.775	0.597	0.819	0.874	0.997	0.897	0.821	0.778
Quick ratio	0.703	0.505	0.646	0.463	0.235	0.393	0.702	0.865	0.794	0.710	0.649
Cash ratio	0.639	0.380	0.560	0.325	0.191	0.290	0.311	0.408	0.267	0.208	0.191

Financial Ratios (Great Wall Motor)

Great Wall Motor	FY24	FY23	FY22	FY21	FY20	FY19	FY18	FY17	FY16	FY15	FY14
Return on equity	7.40%	9.02%	11.87%	9.76%	8.85%	8.95%	12.34%	11.52%	23.46%	21.47%	24.32%
Net operating profit margin	5.37%	2.85%	4.61%	2.98%	3.72%	4.75%	5.52%	4.73%	10.99%	10.18%	12.41%
× Operating asset turnover	7.79	13.04	6.17	185.00	-22.44	6.60	2.12	2.07	2.00	1.92	1.73
= Return on Operating Assets	41.84%	37.19%	28.42%	551.81%	-83.51%	31.33%	11.68%	9.79%	22.02%	19.54%	21.53%
Return on Operating Assets	41.84%	37.19%	28.42%	551.81%	-83.51%	31.33%	11.68%	9.79%	22.02%	19.54%	21.53%
x (Operating Assets/Business Assets)	0.13	0.14	0.24	0.01	-0.06	0.24	0.88	1.00	0.97	0.96	0.99
+ Return on Investment Assets	0.73%	1.68%	2.06%	2.51%	1.57%	0.70%	6.19%	189.53%	15.95%	28.43%	114.87%
x (Investment Assets/Business Assets)	0.87	0.86	0.76	0.99	1.06	0.76	0.12	0.00	0.03	0.04	0.01
= Return on Invested Capital (ROIC)	5.89%	6.71%	8.37%	7.43%	6.81%	8.13%	11.03%	10.62%	21.84%	19.90%	22.42%
Spread	5.73%	6.28%	8.22%	7.22%	6.72%	7.84%	10.93%	10.39%	20.85%	18.39%	21.19%
× Financial leverage	0.26	0.37	0.43	0.32	0.31	0.10	0.12	0.09	0.08	0.09	0.09
= Financial leverage gain	1.51%	2.32%	3.50%	2.33%	2.05%	0.82%	1.31%	0.90%	1.62%	1.57%	1.90%
ROE = Return on Business Assets + Financial leverage gain	7.40%	9.02%	11.87%	9.76%	8.85%	8.95%	12.34%	11.52%	23.46%	21.47%	24.32%
Line items as a percent of sales											
Revenue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Cost of sales	82.96%	84.73%	84.36%	86.95%	85.88%	86.12%	85.77%	84.88%	79.29%	78.58%	75.93%
Selling, general, and admin. expense	6.17%	7.44%	6.03%	6.44%	6.83%	5.72%	5.80%	9.40%	7.85%	9.22%	9.23%
Other operating income/expense	2.24%	0.87%	0.62%	0.77%	0.86%	0.56%	0.18%	0.16%	0.00%	0.00%	0.00%
Net operating profit before tax	5.37%	2.85%	4.61%	2.98%	3.72%	4.75%	5.52%	4.73%	10.99%	10.18%	12.41%
Net non-recurring income/expense	-1.76%	-0.49%	-0.37%	-0.48%	-0.28%	0.35%	-0.02%	-0.47%	-0.34%	-0.08%	-0.09%
Investment and interest income	0.76%	0.87%	1.13%	1.66%	1.40%	0.37%	0.48%	0.51%	0.28%	0.74%	0.75%
Interest expense	0.04%	0.07%	0.03%	0.03%	0.02%	0.02%	0.01%	0.01%	0.04%	0.08%	0.07%
Tax expense	-1.32%	-0.46%	-0.40%	-0.56%	-0.84%	-0.59%	-1.24%	-1.18%	-1.96%	-2.14%	-2.55%
Profit or loss	7.74%	4.05%	6.01%	4.93%	5.19%	4.71%	5.93%	5.61%	11.54%	10.82%	13.08%
Operating expense line items as a percent of revenue (by function) (%)											
Cost of sales	82.96%	84.73%	84.36%	86.95%	85.88%	86.12%	85.77%	84.88%	79.29%	78.58%	75.93%
Selling, general, and administrative expense	6.17%	7.44%	6.03%	6.44%	6.83%	5.72%	5.80%	9.40%	7.85%	9.22%	9.23%
Key profitability ratios (%)											
Gross profit margin	17.040%	15.272%	15.640%	13.052%	14.118%	13.884%	14.231%	15.120%	20.709%	21.416%	24.067%
EBITDA margin	8.345%	3.718%	5.312%	3.856%	4.642%	5.301%	7.087%	6.369%	13.273%	12.301%	14.946%
NOPAT margin	5.371%	2.852%	4.608%	2.983%	3.721%	4.746%	5.519%	4.731%	10.991%	10.179%	12.413%
Net profit margin	7.743%	4.054%	6.009%	4.930%	5.191%	4.709%	5.930%	5.612%	11.540%	10.816%	13.076%
Turnover ratios											
Operating working capital/Sales	-58.07%	-29.69%	-26.97%	-35.04%	-44.98%	-25.70%	7.73%	12.05%	15.64%	13.54%	16.34%
Net non-current assets/Sales	70.91%	37.35%	43.18%	35.58%	40.53%	40.84%	39.50%	36.25%	34.28%	38.53%	41.30%
PP&E/Sales	40.770%	21.620%	26.466%	23.559%	30.767%	33.355%	33.658%	32.220%	29.989%	33.403%	36.019%
Operating working capital turnover	-1.72	-3.37	-3.71	-2.85	-2.22	-3.89	12.94	8.30	6.40	7.38	6.12
Net non-current asset turnover	1.41	2.68	2.32	2.81	2.47	2.45	2.53	2.76	2.92	2.60	2.42
PP&E turnover	2.45	4.63	3.78	4.24	3.25	3.00	2.97	3.10	3.33	2.99	2.78
Trade receivables turnover	8.61	14.13	11.68	14.03	20.79	23.24	24.84	2.01	2.43	2.63	2.32
Days' receivables	41.79	25.48	30.83	25.67	17.31	15.49	14.49	178.90	148.09	137.09	155.23
Inventories turnover	3.42	6.50	6.14	9.77	13.78	15.43	22.32	18.15	16.27	18.46	18.04
Days' inventories	102.45	53.81	57.02	35.84	25.40	22.69	15.68	19.29	21.51	18.96	19.40
Trade payables turnover	1.17	2.14	1.95	1.86	1.78	2.34	2.89	2.70	2.68	2.83	2.61
Days' payables	306.54	168.00	184.47	193.94	202.23	154.08	124.37	133.49	134.31	127.03	138.08
Cash conversion cycle	-162.30	-88.71	-96.61	-132.43	-159.52	-115.90	-94.19	64.70	35.30	29.03	36.56
Liquidity ratios											
Current ratio	1.101	1.070	1.124	1.134	1.225	1.255	1.211	1.177	1.247	1.271	1.351
Quick ratio	0.500	0.494	0.518	0.511	0.300	0.334	0.272	0.941	0.988	1.025	1.162
Cash ratio	0.399	0.383	0.395	0.409	0.239	0.258	0.199	0.087	0.050	0.115	0.130

Appendix 17: Distribution Channels

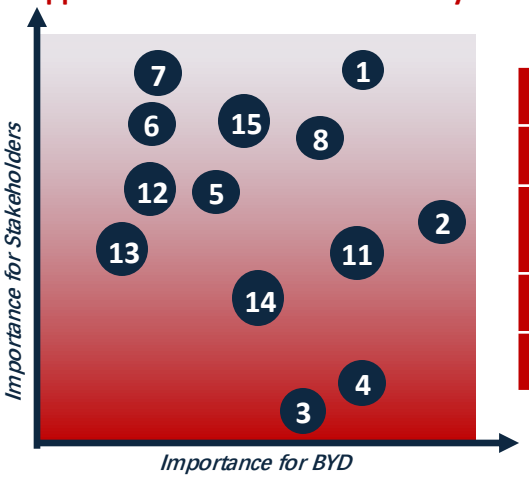
	Dynasty	Ocean	Denza	Formula Leopard	YANGWANG
Dealer 4S store	894	809	/	/	/
Manufacturer's direct store	186	171	514	170	80
City Exhibition Hall	435	341	/	/	/
Supermarket	292	207	/	/	/
Total	1807	1528	514	170	80

Appendix 18: Comparison of Industry Chain Integration between BYD and Competitors

Companies	Raw Material		Power Battery System			Electric Drive System			System Integration
	Raw Material	Electrode Materials	Battery Cell	Battery Module	BMS	Electric Machinery	Electronic Control Hardware (Power Semiconductor)	Electronic Control Software	BMC Integration
Benz	×	×	√	√	√	√	×	√	√
VOLKSWAGEN	√	√	√	√	√	√	×	√	√
Toyota	×	×	×	√	√	×	×	√	√
SAIC Motor	×	×	×	√	√	√	√	√	√
Guangzhou Auto	×	×	√	√	√	×	×	√	√
Great Wall Auto	√	×	√	√	√	√	√	√	√
BYD	√	√	√	√	√	√	√	√	√
Tesla	√	×	√	√	√	√	×	√	√
NIO	×	×	×	√	√	√	×	√	√

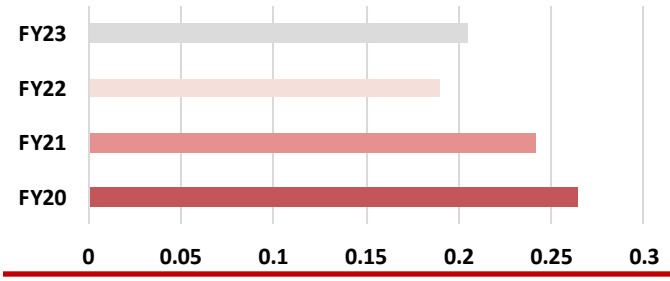
√ Self research and self construction √ Strategic investment/joint venture cooperation x External supply/financial investment

Appendix 19: BYD ESG Issue Priority Matrix

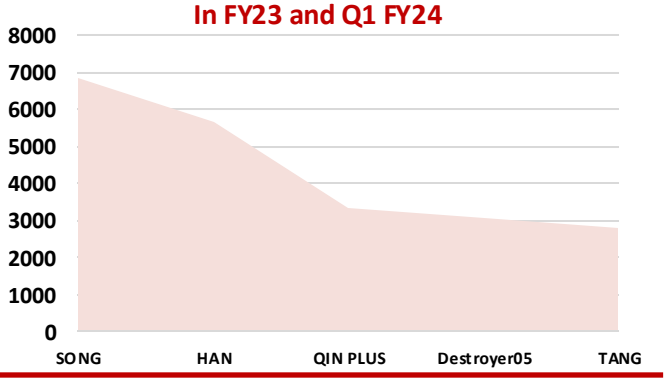


1	Corporate Governance	2	Business Performance	3	Intellectual Property
4	Compliant operation	5	Business Ethics	6	Stakeholder Participation
7	Win-win with partners	8	Sustainable Development	9	Management Supply chain Management
10	Product Liability	11	Technological Innovation	12	Customer Rights
13	Customer Privacy Protection	14	Responsible Marketing	15	Quality Service

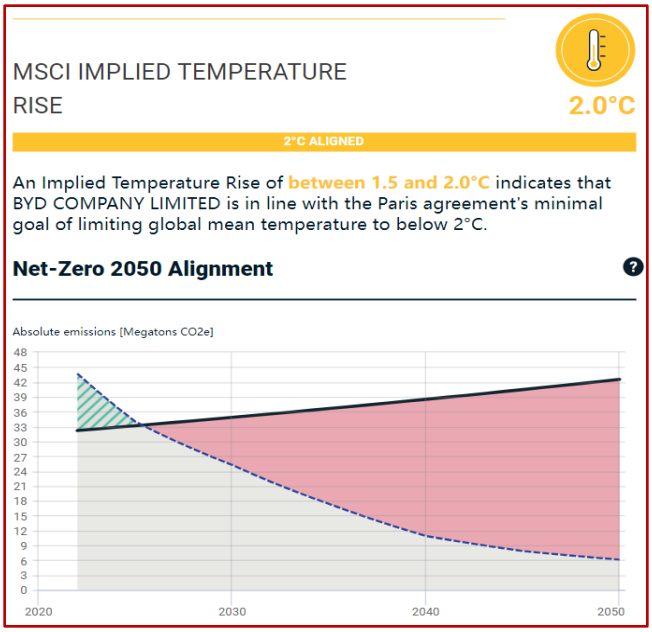
Appendix 20: BYD's Intensity of Green House Gas Emissions (Tons/RMB10,000 revenue)

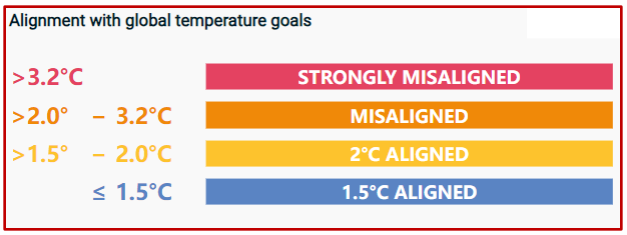


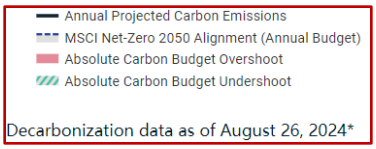
Appendix 21: Top5 Complained Car Series of BYD In FY23 and Q1 FY24



Appendix 22: BYD's Alignment with Global Temperature Goals







Appendix 23: Executives' Biographies

Name and Sex	Position	Tenure	Background
Wang Chuanfu (Male)	Chairman, Executive Director, President	June 10, 2002 to September 19, 2026	In February 1995, he co founded Shenzhen BYD Industrial Co., Ltd. with Mr. Lv Xiangyang and served as the general manager. He is the current Chairman, Executive Director, and President of BYD and is responsible for the general operations of our company and formulating various business strategies.
Lyu Xiangyang (Male)	Vice Chairman and Non-Executive Director	June 10, 2002, to September 19, 2026	He is currently serving as the Vice Chairman and Non-Executive Director of BYD, and holding positions such as Chairman of Rongjie Investment Holdings Group Co., Ltd., Chairman of Rongjie Co., Ltd., and Director of Rongjie Health Technology Co., Ltd.
Xia Zuoquan (Male)	Nonexecutive Director	June 10, 2002 to September 19, 2026	He is currently serving as a non-executive director of BYD and holding positions as chairman of Shenzhen Zhengxuan Investment Co., Ltd., Shenzhen Zhengxuan Qianhai Equity Investment Fund Management Co., Ltd., and Beijing Zhengxuan Investment Co., Ltd.
Cai Hongping (Male)	Independent director	September 8th, 2020 to September 19th, 2026	He is currently an independent non-executive director of BYD, chairman and founding partner of Hande Capital, and serves as the chairman and founding partner of China Eastern Airlines Independent directors of Shanghai Pudong Development Bank Co., Ltd., China Southern Airlines Co., Ltd., and China Merchants Bank Co., Ltd.
Zhang Min (Male)	Independent director	September 8th, 2020 to September 19th, 2027	He is currently the independent non-executive director of BYD, an professor, doctoral supervisor, and head of the accounting department of Renmin University of China School of Business, and serves as China Southern Glass Group Independent Director of Tuan Co., Ltd. and Guotou Capital Co., Ltd
Yu Ling (Female)	Independent director	September 19, 2023 to September 19, 2026	She is currently an independent non-executive director of BYD and serves as a member of the China Economic Law Research Association Director of the Financial and Tax Law Research Association, Case Law Research Association, and Asian Competition Law Association.

Appendix 24: References

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